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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
CV	Curriculum Vitae
HR	Human Resources
JWT	JSON Web Token
OCR	Optical Character Recognition
PDF	Portable Document Format
REST	Representational State Transfer
UI	User Interface
UML	Unified Modeling Language

GENERAL INTRODUCTION

Organizational practices are now dependent not only on human resources but also on the ability to manage processes effectively and adapt to rapid change driven by the evolution of digital technologies. In this context, recruiting is strategically important, as it directly impacts team building and organizational success. However, the complexity of this process has increased because of the growing number of applications and the demand for faster, more accurate treatment. Consequently, organizations are turning to digital solutions that centralize information, improve workflows, and support decision-making, turning recruitment into a structured process that requires efficient tools.

In this setting, Sopra HR Software is recognized as a leading player in human resource management solutions. Operating in a field where technological innovation and service quality are essential, the company emphasizes the need to improve its internal processes and business tools. In this sphere, the digitalization of recruitment is a major lever for streamlining application processing, enhancing the user experience, and supporting recruiters in their day-to-day activities.

The implementation of an intelligent recruitment portal aims to meet both operational and strategic needs by providing a more cohesive framework for managing job offers, applications, and interactions among all stakeholders involved in recruitment. Consequently, a central question arises : how can a recruitment portal be designed to centralize applications, facilitate CV processing, and effectively support recruiters in their decision-making, while aligning with a logic of continuous improvement in the recruitment process ?

To explore the identified problem and present the work conducted during the internship, the report is structured into six chapters :

- **Chapter One** presents the general framework of the project by introducing the host organization and situating the project within its context. It analyzes the existing recruitment process and its limitations, compares recruitment solutions, and justifies the proposed intelligent recruitment portal.
- **Chapter Two** addresses requirements analysis and specification. It describes the functional and non-functional requirements, identifies the system actors, presents the product backlog and sprint planning, and introduces the working environment and technologies used.

- **Chapter Three** covers Sprint 1, focusing on authentication and initial setup, including user registration, sign-in, and role and permission management.
- **Chapter Four** covers Sprint 2, dedicated to recruitment management, encompassing job offer lifecycle, application submission, pipeline management, and manager offer recommendations.
- **Chapter Five** covers Sprint 3, which introduces AI-powered CV analysis and scoring to support automated candidate evaluation and ranking.
- **Chapter Six** covers Sprint 4, addressing finalization and deployment, including interview planning and scheduling features.

Finally, the report concludes with a summary that recaps the context of the work, reviews the project, and presents possible directions for future enhancements.

Chapter

1

GENERAL PROJECT CONTEXT

Introduction

This chapter establishes the project's general framework by introducing the host organization and situating it within its institutional and operational context. It examines the existing recruitment process at Sopra HR Software and highlights its limitations through detailed analysis. A comparative study of available recruitment solutions is then presented to identify gaps in current approaches. Finally, the chapter justifies the proposed intelligent recruitment portal as a response to the identified needs and challenges.

1.1 Presentation of the Host Organization

Sopra Steria is a well-known European group recognized for its expertise in consulting, digital services, and business software publishing, particularly in human resources and banking. In 2024, the Sopra Steria Group operates in nearly 30 countries and employs around 51,000 people. Sopra HR Software is a subsidiary of the Sopra Steria Group, specializing in Human Resources management, payroll administration, and talent management, both nationally and internationally. More than 850 clients use its solutions across over 54 countries. Designed to meet the specific needs of HR departments, these solutions are intended for large and medium-sized enterprises in both the public and private sectors.

FIGURE 1.1 – Logo of Sopra HR Software

1.2 Project Study

This section outlines the project context and analyzes the existing system to identify its limitations. It also presents a comparative study of current recruitment portals, highlighting their strengths and shortcomings relative to the proposed platform. This analysis supports the definition and justification of the proposed solution.

1.2.1 Project Context

This project was carried out as part of a Final Year Project at the Higher Institute of Technological Studies of Nabeul, to apply academic knowledge to the development of an innovative software solution. It focuses on the design and implementation of an intelligent recruitment portal intended to support and modernize the recruitment processes of Sopra HR Software. In today's digital environment, recruitment platforms play a key role in streamlining hiring processes and improving interaction between candidates and recruiters. Accordingly, this project aims to design a centralized and intelligent recruitment platform tailored to the needs of Sopra HR Software, leveraging modern technologies to enhance efficiency and user experience.

1.2.2 Analysis of the Existing System

The recruitment process in many organizations, including Sopra HR Software, relies on manual procedures and multiple disconnected tools, where job offers and applications are managed across various platforms. As a result, recruiters must manually handle tasks such as CV screening, shortlisting, and interview coordination. As highlighted in studies on online recruitment systems, handling large volumes of applications without an automated applicant tracking system makes it difficult to identify suitable candidates efficiently and consistently. This fragmentation slows down the recruitment process and increases the workload on HR teams.

Current Recruitment Practices at Sopra HR Software

Several limitations characterize the existing recruitment approach :

- **Lack of centralization** : Candidate data and job offers are spread across multiple platforms.
- **Time-consuming CV screening** : Manual sorting of resumes increases processing time and recruiter workload.
- **Reduced decision accuracy** : Comparing candidates becomes difficult without structured evaluation tools.
- **Limited process visibility** : Recruiters struggle to track application status and recruitment stages effectively.

These limitations reduce the overall efficiency of recruitment activities and prevent recruiters from focusing on strategic HR tasks.

1.2.3 Comparative Study of Existing Recruitment Solutions

This comparative study analyzes existing recruitment platforms as reference systems to identify best practices and functional limitations. Its objective is to highlight the gaps between generic recruitment solutions and the specific needs of an internal recruitment portal dedicated to Sopra HR Software Tunisia.

Aspect	Public Recruitment Platforms (LinkedIn, Indeed, etc.)	ATS-Oriented Tools (Workable)	Proposed Internal Portal (Sopra Recruit)
Target Scope	Multi-Company, Global	Multi-Company	Single Company (Sopra HR Software)
Recruitment Context	Generic	Semi-Customizable	Fully context-specific
Job Offer Management	Public and standardized	Centralized	Dedicated to internal positions
Candidate Flow	High volume, external candidates	Controlled pipelines	Targeted candidates
Customization	Limited	Moderate	High
Adaptation to Local Process	Low	Moderate	Complete
Data Ownership	Platform-dependent	Platform-dependent	Fully internal
Recruitment Monitoring	Standard statistics	Advanced pipelines	Tailored dashboards
Main Limitations	Lack of customization	Cost and rigidity	Scope limited to one company

TABLE 1.1 – Comparative Overview of Recruitment Solutions

The comparison shows that public recruitment platforms are primarily generic and insufficiently adapted to company-specific processes, while ATS tools remain constrained by predefined workflows. In contrast, the proposed internal portal is tailored to Sopra HR Software, enabling better integration, customization, and data control.

1.2.4 Proposed Solution

To address these limitations, this project proposes designing and developing an intelligent recruitment portal for Sopra HR Software. The solution is built around five core capabilities :

- A centralized platform for managing job offers, applications, and recruitment stages from a single interface
- Automatic CV parsing that extracts and structures candidate information, including skills, experience, education, languages, certifications, and soft skills
- An AI-based matching engine that scores each candidate against offer requirements using configurable weighted criteria
- A collaborative recruitment workflow enabling HR officers and managers to evaluate, forward, and validate applications with full traceability
- Real-time notifications, interview scheduling, and a recruitment dashboard with exportable statistics

This portal allows Sopra HR Software to reduce manual effort, accelerate hiring decisions, and improve the overall quality and consistency of its recruitment process.

1.3 Working Methodology Choice

During the development of an information system, it is essential to adopt a clear, structured methodology to guide the project through its different stages. A development methodology defines the processes and steps required to organize the work, ensure progress at each phase, and align the system being developed with user needs. For this project, the Agile methodology was chosen, as it provides flexibility, encourages continuous improvement, and supports incremental delivery throughout the development cycle.

1.3.1 Agile Method

Agile is a software development methodology widely adopted for its ability to handle complexity and evolving requirements. According to the Agile Manifesto [1], it is based on an iterative and incremental approach that prioritizes individuals and interactions, working software, and responsiveness to change over rigid planning and processes. Development is organized into short cycles, each delivering a functional increment that stakeholders can review and provide feedback on. As described by Atlassian [2], this structure improves transparency, product quality, and team adaptability throughout the project lifecycle. Among the various Agile frameworks, SCRUM was selected for this project and is presented in the following section.

1.3.2 SCRUM Methodology

Scrum is a lightweight Agile framework widely used to manage and develop complex systems in an iterative, incremental manner. According to the Scrum Guide, Scrum structures development activities into fixed-length iterations called sprints, which typically last from one to four weeks. Each sprint is executed with a clear objective and results in the delivery of a potentially usable product increment, allowing continuous inspection, adaptation, and value creation throughout the project lifecycle [2].

Scrum applies Agile principles by emphasizing collaboration, frequent delivery, and responsiveness to change. It is founded on empiricism, which relies on transparency, inspection, and adaptation as core pillars, enabling teams to progressively refine the product while responding effectively to evolving requirements [2].

1.3.3 SCRUM Actors

The Scrum framework defines three key roles, each with specific responsibilities [2] :

- **Product Owner** : Responsible for defining the product vision and representing stakeholder needs. The Product Owner manages and prioritizes the product backlog to maximize the value delivered by the development team.
- **Scrum Master** : Ensures that the Scrum framework is properly understood and applied. The Scrum Master facilitates communication, removes impediments, and promotes adherence to Agile principles.
- **Development Team** : A self-organized and cross-functional team, typically composed of 3 to 9 members, responsible for designing, developing, and delivering functional increments at the end of each sprint.

The structured yet flexible framework enables effective collaboration, continuous improvement, and high-quality delivery throughout the development process.

Chapter

2

REQUIREMENTS SPECIFICATION

Plan

Introduction

This chapter is dedicated to the project's preparation phase. It begins with the elicitation and specification of system requirements, then presents the functional and non-functional requirements of the system. The chapter also describes the organization and management of the project using the SCRUM methodology, including team roles, the product backlog, and sprint planning. It then presents the working environment in terms of hardware, tools, and technologies used during development, and concludes with the requirements modeling through a global use case diagram.

2.1 Requirements Elicitation

The purpose of requirements elicitation is to identify the stakeholders' needs and expectations in the recruitment process. In this project, requirements were gathered through an analysis of the existing recruitment system at Sopra HR Software and discussions with recruiters to understand their difficulties and daily practices. This step highlighted the need for a centralized recruitment platform, better candidate tracking, and improved support for recruiter decision-making, which serve as the basis for the formal requirements defined in the next section.

2.1.1 Requirements Specifications

Requirements specification is the process of formally defining what the system must do and how it must behave. Based on the information gathered in the previous section, both functional and non-functional requirements are identified and expressed through clear textual descriptions and structured models, providing a precise foundation for the design and development phases.

2.1.2 Identification of Actors

An information system interacts with different users, each having specific roles and responsibilities. In the proposed recruitment portal, these users are identified as system actors and are presented in Table 2.1.

Actor	Description
Administrator / HR	Manages the recruitment platform by creating job offers, configuring scoring criteria, and supervising applications and system settings.
Manager	Reviews applications forwarded by the HR department, contributes to candidate evaluation and final selection, and can submit job offer recommendations to the HR officer when a position needs to be filled.
Candidate	Applies for job offers by submitting a CV and tracks the status of submitted applications.
Visitor	Browses job offers without authentication and registers to become a candidate.

TABLE 2.1 – Identification of System Actors

2.2 Functional and Non-Functional Requirements

This section defines the functional and non-functional requirements of Sopra Recruit. Functional requirements describe what the system must do, including offer management, application processing, CV analysis, scoring, and user management. Non-functional requirements define the quality constraints the system must satisfy, including security, performance, usability, and maintainability.

2.2.1 Functional Requirements

The functional requirements are organized by actor. Each actor, including the Visitor, Candidate, Human Resources Officer, and Manager, is associated with a specific set of interactions and capabilities within the platform.

Visitor (Unauthenticated User)

The Visitor represents a user who accesses the platform without authentication. The functional needs associated with this actor are as follows :

- Browse the institutional pages of the recruitment portal.
- Consult the list of published job offers.

- Search and filter job offers according to defined criteria.
- View detailed information about a selected job offer.
- Create a candidate account through a public registration form.
- Activate the account by confirming the email address.

Candidate

The Candidate is an authenticated user who applies for job offers and manages personal applications. The Candidate's functional requirements are :

- Sign in and sign out of the platform securely.
- Manage personal account information and update profile data.
- Consult available job offers and their detailed descriptions.
- Apply to job offers by submitting a CV and complementary information.
- Reuse an existing CV or upload a new one during application submission.
- Track the status of submitted applications.
- Consult planned interviews and receive related notifications.

Human Resources Officer (HR) / Administrator

The Human Resources Officer acts as the administrator of the recruitment platform and is responsible for managing and supervising the entire recruitment process. This role has full access to the system and oversees both functional and organizational aspects of the application. The functional requirements for this role include :

- Access the secured administration interface.
- Manage user accounts and assign roles and permissions.
- Create, publish, update and archive job offers.
- Consult, filter, and manage received applications according to their status, score, or submission date.
- Evaluate applications and decide on rejection or progression.
- Forward selected applications to managers.
- Plan and manage recruitment interviews.
- Review and modify job offers proposed by managers, and decide whether to approve or reject them, with automatic publication upon approval.
- Monitor recruitment activity through dashboards and reports.

Manager

The Manager is an authenticated internal user who participates in the recruitment decision-making process within a specific department. The platform may involve multiple managers, each responsible for evaluating candidates and proposing recruitment needs related to their organizational unit. The functional requirements associated with this actor include :

- Access the secured management interface.
- Consult the applications forwarded by the HR department.
- Review candidate profiles and evaluation results.
- Validate or reject candidate applications.
- Participate in interview follow-up.
- Manage personal notifications and profile information.
- Propose a new job offer by submitting a dedicated request form when a position becomes vacant or requires reinforcement.
- Submit the proposed job offer to the Human Resources Officer for review and validation.

2.2.2 Non-Functional Requirements

Beyond the functionalities described above, the recruitment portal must meet a set of quality requirements that determine its acceptance by end users (candidates, HR officers, managers) and the platform's long-term sustainability. These requirements are formulated as observable constraints rather than implementation choices.

Security and Confidentiality

- Users must be logged in to access protected parts of the system.
- Users can only perform actions allowed by their role (Candidate, HR, or Manager).
- Passwords and sensitive personal data (such as CVs and contact details) must be stored securely.
- A candidate account is activated only after the email address is confirmed within a limited time.
- All communication between front-end, back-end, and AI service must be secure.
- The platform must respect data-protection rules, including user consent, access to personal data, and controlled storage of CVs.

Performance and Responsiveness

- The main pages (dashboards, application lists, calendars) should remain smooth and responsive, with loading times under 2 seconds under normal use.
- CV analysis and scoring should be completed quickly, with a target processing time of less than 10 seconds per application.
- AI processing should run in the background so that candidates can submit their applications without waiting.

Reliability and Availability

- The platform should be available at least 99% of the time during business hours.
- If the AI service is temporarily unavailable, applications must still be saved, and scoring can be recalculated later.
- The recruitment process must remain consistent, and important errors must be recorded instead of being ignored.

Usability and Ergonomics

- The interface should have a modern and professional design, and each screen should follow the natural way users work.
- Every action performed by a user should immediately show visual feedback, and the system status should be displayed consistently across the application.

Compatibility and Accessibility

- The platform should be accessible through major web browsers (Chrome, Edge, Firefox, Safari) without requiring any installation.
- The interface should work well on desktop, laptop, and tablet screens, use consistent language, and allow adding other languages in the future.

Scalability

- The platform should support a growing number of users, job offers, and applications without noticeable slowdowns, and its main features should remain independent.

Maintainability and Evolvability

- The source code should be well-organized into clear modules and documented so that a new developer can easily understand and maintain the project.
- Configuration settings (database, AI service, email server, security keys) should be kept separate from the code so the application can run in different environments.
- HR users should be able to adjust the recruitment workflow and scoring criteria without needing a new software update.

Portability and Deployment

- The platform's components should be deployable separately, for example, on cloud or container-based systems, without changing the application logic.

2.3 Project Management Using SCRUM

The Agile SCRUM framework, presented in Chapter 1, was used to manage the development of this project. Its iterative structure allowed the team to organize work into four sprints, prioritize features progressively, and deliver functional increments at regular intervals. The following sections present the team composition, the product backlog, and the sprint planning adopted for this project.

2.3.1 Team and Roles

The project team was organized according to the three SCRUM roles defined in Chapter 1. Table 2.2 presents each role, the person responsible, and their specific involvement in this project.

Scrum Role	Assigned Member	Main Responsibilities
Product Owner	Mr. Ramzi Ben Chamekh	Defines and prioritizes functional requirements, validates deliverables at each sprint review.
Scrum Master	Ms. Nada Bouhlel	Ensures SCRUM practices are followed, facilitates sprint ceremonies, supports the development team.
Development Team	Mayssem ESSMAII & Bechir JABLI	Designs, implements, and tests all platform features across the four sprints.

TABLE 2.2 – SCRUM Team Composition

2.3.2 Product Backlog

The product backlog constitutes the ordered list of all features and requirements to be implemented in the intelligent recruitment portal. Each item is expressed as a user story and is assigned a priority level and a story point estimate. The backlog is organized into thirteen functional epics covering all major system capabilities [10].

Epic	Actor	ID	User Story	Priority	SP
Epic 1 Authentication & Registration	Visiteur	US-AUTH-01	As a visitor, I want to register with my name, email and password so that I can create a candidate account on the platform.	High	3
	Candidat	US-AUTH-02	As a candidate, I want to verify my account via an activation email so that my identity is confirmed before I can sign in.	High	2
	Candidat	US-AUTH-03	As a candidate, I want to sign in with my email and password so that I can access my personal space.	High	2
	Candidat	US-AUTH-04	As a candidate, I want to sign out so that my session is closed securely.	High	1
Epic 2 User Management	RH	US-USR-01	As an RH, I want to create user accounts with an assigned role so that new members can access the platform immediately.	High	3

Epic	Actor	ID	User Story	Priority	SP
	RH	US-USR-02	As an RH, I want to list, search and update user information so that records remain accurate.	High	3
	RH	US-USR-03	As an RH, I want to change a user's role so that each person accesses only the features relevant to their responsibility.	High	2
Epic 3 User Profile	Candidat / RH	US-PROF-01	As a user, I want to view and update my profile (phone, address, photo) so that my information stays up to date and visible to others.	Medium	2
Epic 4 Job Offer Management	RH	US-OFF-01	As an RH, I want to create a job offer with title, description, required skills and contract type so that candidates can discover available positions.	High	5
	RH	US-OFF-02	As an RH, I want to change the status of an offer (draft, published, archived) so that only relevant positions are visible to candidates.	High	2
	RH	US-OFF-03	As an RH, I want to view all offers regardless of status so that I have a complete view of the recruitment pipeline.	High	2
	Visiteur	US-OFF-04	As a visitor, I want to browse published offers and filter them by keyword, experience and contract type so that I can find relevant opportunities.	High	3
Epic 5 Application Management	Candidat	US-CAND-01	As a candidate, I want to apply to an offer by uploading my CV and cover letter so that my application is registered.	High	5
	Candidat	US-CAND-02	As a candidate, I want to view the list and current status of my applications so that I stay informed about my progress.	High	2
	RH	US-CAND-03	As an RH, I want to view and filter all applications by status, score or date so that I can manage the recruitment pipeline efficiently.	High	3
	RH	US-CAND-04	As an RH, I want to change an application status (under evaluation, forwarded to manager, rejected) so that qualified profiles advance in the workflow.	High	3

Epic	Actor	ID	User Story	Priority	SP
	Manager	US-CAND-05	As a manager, I want to view applications forwarded to me and validate or reject them so that I participate in the final selection.	High	3
	RH	US-CAND-06	As an RH, I want to download a candidate's CV so that I can review the original document.	Medium	1
Epic 6 CV & AI Analysis	RH	US-CV-01	As an RH, I want to view the structured data automatically extracted from a CV (skills, experience, education, languages) so that I can evaluate a candidate quickly.	High	5
	RH	US-CV-02	As an RH, I want the system to extract certifications and soft skills from a CV so that the candidate profile is as complete as possible.	Medium	3
Epic 7 AI Scoring	RH	US-SCORE-01	As an RH, I want to define weighted scoring criteria for each offer so that candidates are evaluated objectively against the position requirements.	High	5
	RH	US-SCORE-02	As an RH, I want to view the AI matching score and its breakdown per criterion so that I can make informed selection decisions.	High	3
	RH	US-SCORE-03	As an RH, I want to sort applications by score and trigger a recalculation so that the ranking always reflects the latest criteria.	Medium	2
Epic 8 Interviews & Calendar	RH	US-ENT-01	As an RH, I want to schedule an interview with a date, location and format (in-person, video, phone) so that the process advances in an organised manner.	High	5
	RH	US-ENT-02	As an RH, I want to confirm or cancel a planned interview so that the calendar stays accurate.	Medium	2
	Manager	US-ENT-03	As a manager, I want to view the interview calendar for my team so that I can avoid scheduling conflicts.	High	3
	Candidat	US-ENT-04	As a candidate, I want to view my upcoming interviews with their date, location and format so that I can prepare accordingly.	High	2

Epic	Actor	ID	User Story	Priority	SP
Epic 9 Evaluations	RH / Manager	US-EVAL-01	As an RH or manager, I want to evaluate an application with a structured comment and decision so that my assessment is formally recorded and traceable.	High	3
	RH	US-EVAL-03	As an RH, I want to view the full evaluation history of an application so that all assessments are visible and traceable.	Medium	2
Epic 10 Offer Recommendation	Manager	US-REC-01	As a manager, I want to submit a job offer recommendation to the RH so that recruitment needs are communicated promptly.	Medium	3
	RH	US-REC-02	As an RH, I want to review, accept or reject a manager's recommendation so that I retain control over published offers.	Medium	3
Epic 11 Notifications	Candidat / RH / Manager	US-NOTIF-01	As an authenticated user, I want to receive notifications for relevant recruitment events so that I stay informed without checking the platform manually.	High	2
Epic 12 Dashboard & Statistics	RH	US-DASH-01	As an RH, I want to view a dashboard with key indicators (total applications, published offers, selection rate) so that I can monitor recruitment activity in real time.	High	5
	RH	US-DASH-02	As an RH, I want to view detailed statistics (applications per month, top offers, average time-to-hire) and export a report so that I can share results with management.	Medium	5
Epic 13 Public Site	Visiteur	US-PUB-01	As a visitor, I want to browse the landing page so that I can discover Sopra HR and its recruitment opportunities.	Medium	2
	Visiteur	US-PUB-02	As a visitor, I want to access institutional pages (About us, Our professions, Our sectors, Culture) so that I can learn about the company before applying.	Medium	3

TABLE 2.3 – Product Backlog — Intelligent Recruitment Portal (Sopra HR)

2.3.3 Sprint Planning

After the definition of the product backlog, the workload is distributed across the planned sprints. This distribution is presented in Table 2.4, which outlines the main activities and objectives of each sprint.

Sprint	Main Tasks	Duration
Sprint 1 — Authentication & Setup	Registration with email verification ; sign in and sign out for all roles ; user creation and role assignment ; profile management ; landing page and institutional pages.	4 weeks
Sprint 2 — Recruitment Management	Job offer creation, publication and archiving ; offer browsing with filters ; application submission with CV and cover letter ; application pipeline management ; manager offer recommendations.	4 weeks
Sprint 3 — Artificial Intelligence	Automatic CV extraction (skills, experience, education, languages) ; structured candidate profile view ; weighted scoring criteria configuration ; AI matching score with breakdown ; score-based ranking and recalculation.	4 weeks
Sprint 4 — Finalization & Deployment	Interview scheduling and calendar management ; structured evaluation with comment and decision ; evaluation history per application ; notifications for recruitment events ; recruitment dashboard and exportable statistics.	4 weeks

TABLE 2.4 – Sprint Planning

2.4 Working Environment

This section describes the working environment used throughout the development of the intelligent recruitment portal. It covers the hardware configuration, the software tools, and the technologies selected to implement the various components of the system.

2.4.1 Hardware Environment

Component	Specification
Model	HP EliteBook 840 G8
RAM	16GB
CPU	11th Gen Intel Core i5-1145G7 2.60 GHz
GPU	Intel UHD Graphics
Operating system	Windows 11

TABLE 2.5 – Hardware Environment

2.4.2 Software Environment

Operating System : Windows 11 is the operating system installed on the computer used to build this project. An operating system is the main software that manages the computer and allows other programs to run on it. Windows 11 offers a clean and modern working space that works well with many development tools, making it easy to write, test, and run the application without any problems.

FIGURE 2.1 – Windows 11 Logo

2.4.3 Development Tools

Visual Studio Code : Visual Studio Code, also called VS Code, is a free and lightweight code editor made by Microsoft. It was used to write and edit the frontend code of this project. VS Code is very popular among developers because it is simple to use, starts up quickly, and supports many programming languages. It also has a large collection of extensions, which are small add-ons that add useful features like code suggestions, error detection, and tools for working with Git [3].

FIGURE 2.2 – Visual Studio Code Logo

IntelliJ IDEA : IntelliJ IDEA is a professional code editor made by JetBrains, designed mainly for writing Java applications. It was used to build the backend of this project. It helps developers write code faster by offering smart suggestions, automatic error detection, and built-in tools for running and testing the application. It is widely used in the industry for building large and reliable backend systems.

FIGURE 2.3 – IntelliJ IDEA Logo

Overleaf : Overleaf is a free online tool used to write documents using LaTeX, which is a system that formats text in a professional and structured way. It was used to write this project report. Overleaf is easy to use because it works directly in the browser without needing to install anything on the computer, and it shows a live preview of the document while editing.

FIGURE 2.4 – Overleaf Logo

Git : Git is a free tool that helps developers keep track of all the changes made to the code over time. It is called a version control system. With Git, it is possible to save different versions of the project, go back to an older version if something goes wrong, and work with a team without accidentally overwriting each other's work.

FIGURE 2.5 – Git Logo

StarUML : StarUML is a diagram drawing tool used to design and show the structure of a software system. In this project, it was used to draw UML diagrams, which are special diagrams that show how different parts of the application are connected and how they work together. These diagrams make it easier to understand and explain the system before writing any code.

FIGURE 2.6 – StarUML Logo

Swagger : Swagger is a tool used to document and test the API of the backend application. An API is the way the frontend and backend talk to each other. Swagger creates an interactive web page where it is possible to see all the available API routes and send test requests directly from the browser, making it easy to check that the backend is working correctly [14].

FIGURE 2.7 – Swagger Logo

2.4.4 Technologies Used

React JS : React JS is a free and open-source JavaScript library created by Facebook. It is used to build the user interface of the web application. React makes it easy to create interactive pages by dividing the screen into small, reusable parts called components. Each component is responsible for one part of the page, which makes the code cleaner, easier to read, and easier to update [5].

FIGURE 2.8 – React JS Logo

JavaScript : JavaScript is a programming language that runs directly in the web browser. It is used to make web pages interactive and dynamic. In this project, JavaScript is the main language used for the frontend, since React JS is built on top of it. It allows the application to respond to user actions like clicking buttons, filling forms, and loading new data without refreshing the whole page [6].

FIGURE 2.9 – JavaScript Logo

CSS3 : CSS3, which stands for Cascading Style Sheets version 3, is a language used to control how web pages look. It sets colors, fonts, sizes, spacing, and the layout of elements on the screen. CSS3 also includes tools for making the interface responsive, which means the page automatically adjusts its layout to look good on different screen sizes, such as phones, tablets, and computers.

FIGURE 2.10 – CSS3 Logo

Vite : Vite is a modern build tool used during the development of the frontend. A build tool helps by automatically turning the source code into a format that the browser can understand and run. Vite is known for being very fast, which means the browser refreshes almost instantly every time a change is made to the code. This makes the development process much faster and more comfortable.

FIGURE 2.11 – Vite Logo

Spring Boot : Spring Boot is a Java framework used to build the backend of the application. A framework is a set of ready-made tools and rules that make it easier to develop software. Spring Boot makes it simple to create backend services that receive requests from the frontend, apply the business logic, and communicate with the database. It is widely used in professional projects because it is reliable, fast to set up, and easy to extend [4].

FIGURE 2.12 – Spring Boot Logo

JDK : The JDK, or Java Development Kit, is a set of tools needed to write, compile, and run Java programs. Compiling means turning the code written by the developer into a format that the computer can run. The JDK includes everything needed to develop Java applications, and it is required to use Spring Boot on the computer.

FIGURE 2.13 – JDK Logo

Python : Python is a simple and easy-to-read programming language that is widely used for artificial intelligence and data processing tasks. In this project, Python is used for the AI module, which is responsible for reading and analyzing CVs. It extracts useful information from uploaded CV files and calculates a score based on how well the candidate matches the job requirements [8].

FIGURE 2.14 – Python Logo

Node.js : Node.js is a tool that allows JavaScript code to run outside of the web browser, directly on the computer or server. Normally, JavaScript only works inside a browser, but with Node.js it can also be used for other tasks. In this project, Node.js is mainly used as part of the frontend development environment, since tools like npm, which is used to install JavaScript packages, need it to work.

FIGURE 2.15 – Node.js Logo

PostgreSQL : PostgreSQL is a free and open-source database management system used to store all the data of the application. A database is like a large and organized storage space where all information, such as user accounts, job offers, and candidate applications, is saved. PostgreSQL is reliable and supports many types of data operations, making it a good choice for web applications that need to store and manage large amounts of data [7].

FIGURE 2.16 – PostgreSQL Logo

2.5 Requirements Modeling

Requirements modeling provides a structured and visual representation of the system's expected behavior and the interactions between its different actors. In this project, the Unified Modeling Language (UML) is used to describe these interactions through use case diagrams, which serve as the primary tool for capturing and communicating functional requirements at a high level of abstraction.

2.5.1 Global Use Case Diagram

Use case diagrams are used to show the expected behavior of a system from the users' point of view. They help identify and organize the functional requirements by showing the interactions between users (actors) and the different system features. The system objectives must be clearly defined, and its development should be guided by the users' needs [11].

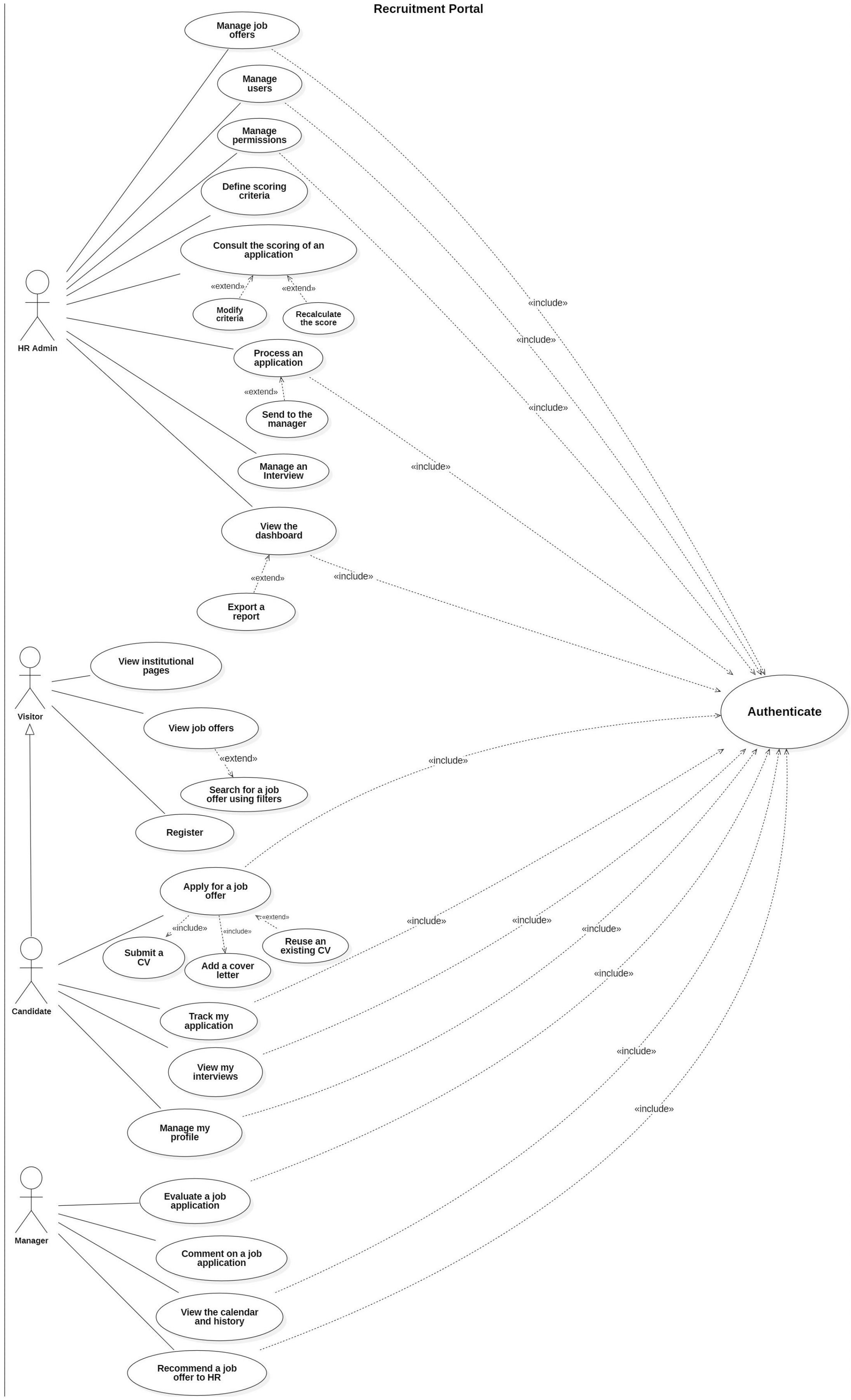

FIGURE 2.17 – Global Use Case Diagram

Conclusion

This chapter explained the preparation phase of the intelligent recruitment portal project. First, we identified the system users and defined both functional and non-functional requirements. Then, we described the SCRUM method used to manage the project, including team roles, the product backlog, and sprint planning. We also presented the working environment, including the hardware, software, and tools used during development. Finally, we showed the global use case diagram to give a clear overview of how users interact with the system. This preparation provides a strong base for the design and implementation chapters that come next.

Chapter

3

SPRINT 1 : AUTHENTICATION AND SETUP

Plan

Introduction

Sprint 1 represents the foundational phase of the Sopra Recruit platform. It covers the implementation of the complete authentication system, including visitor registration with email verification and secure sign-in and sign-out mechanisms for all platform roles. It also encompasses user creation with role assignment managed by the HR administrator, a profile management module available to all authenticated users, and the public-facing interface composed of the landing page and institutional pages. These features form the essential entry point of the platform and constitute the prerequisite layer for all subsequent development sprints.

3.1 Sprint Backlog

Table 3.1 presents the sprint backlog for Sprint 1, detailing the user stories, their estimated duration, the associated implementation tasks, and the difficulty level assigned to each item.

User Story	Duration	Task	Difficulty
Registration with email verification	1 week	Implement the registration form ; create the backend user creation endpoint ; integrate the email verification service ; handle account activation via confirmation link.	Medium
Sign in and sign out for all roles	1 week	Develop the JWT-based authentication endpoint ; implement role-based redirection after login ; manage refresh tokens ; ensure secure session termination on sign-out.	Medium
User creation and role assignment	1 week	Build the HR administration interface for user creation ; implement role assignment and permission management ; allow the HR officer to activate and manage platform accounts.	High
Profile management	3 days	Develop the profile view and edit form ; integrate profile picture upload ; enable personal information update for all authenticated roles.	Low
Landing page and institutional pages	4 days	Design and implement the public landing page presenting Sopra HR ; create the institutional pages (About Us, Our Professions, Our Sectors, Culture).	Low

TABLE 3.1 – Sprint 1 Backlog

3.2 Sprint Objective

The primary objective of Sprint 1 is to deliver a fully operational authentication and user management system that serves as the structural backbone of the Sopra Recruit platform. By the end of this sprint, visitors will be able to register on the platform and activate their accounts through an email verification mechanism, while all platform roles — candidate, HR officer, and manager — will have access to secure sign-in and sign-out functionality. The HR administrator will be able to create user accounts, assign roles, and manage platform access accordingly. A profile management module and a polished public interface, including the landing page and institutional pages, will also be delivered, ensuring the platform is both accessible to the public and ready for authenticated usage in the sprints that follow.

3.3 Analysis

3.3.1 Use Case Diagram

A use case diagram is a UML diagram that illustrates the interactions between users (called actors) and the system. It describes the different functionalities (use cases) that the system offers to its users, showing who uses what and how, without going into technical details [11].

The use case diagram for Sprint 1, presented in Figure 3.1, identifies the interactions between the four system actors — Visitor, Candidate, HR Officer, and Manager — and the authentication and setup features delivered during this sprint.

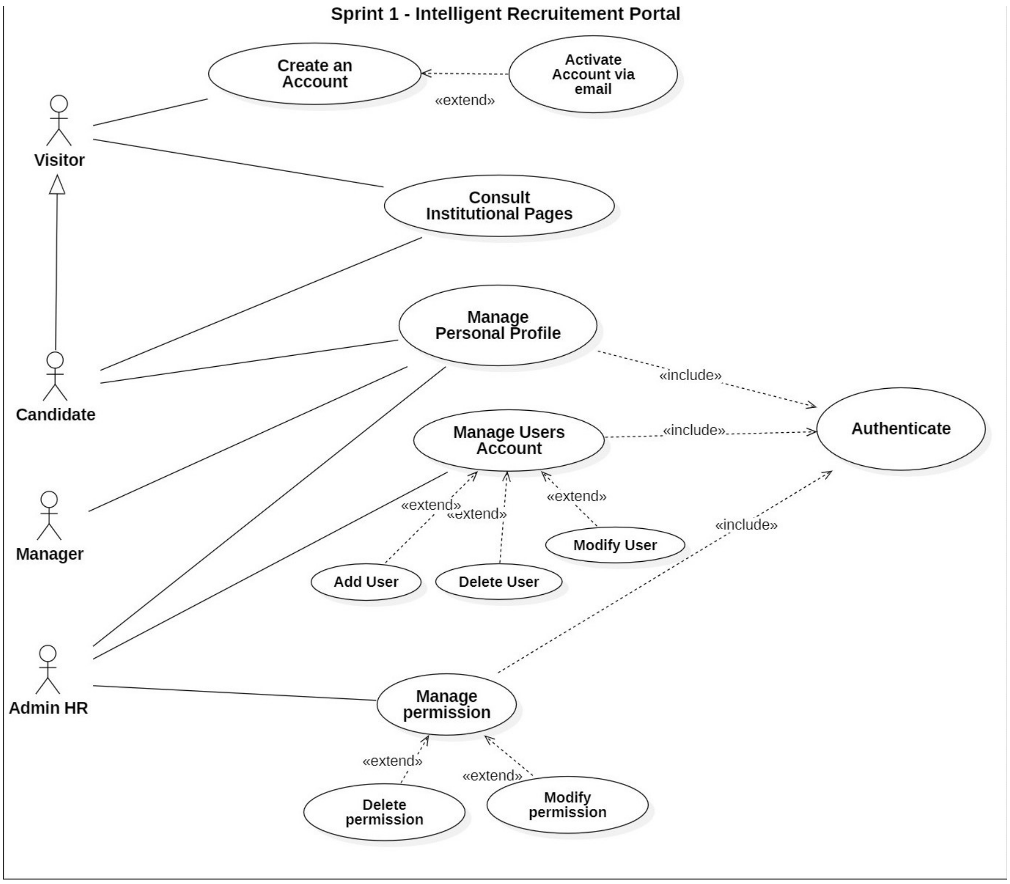

FIGURE 3.1 – Use Case Diagram — Sprint 1 : Authentication and Setup

3.3.2 Textual Description of Use Case

This section provides a textual description of the main use cases appearing in the Sprint 1 use case diagram. Each use case is described through its initiating actor, goal, pre-conditions, post-conditions, main success scenario, and alternative scenarios.

3.3.2.1 Use Case Description : “Register”

Use Case : Register	
Use Case	Register
Initiating Actor	Visitor
Goal	Allow a visitor to create a new candidate account on the Sopra Recruit platform by providing their personal information and confirming their email address.
Pre-condition	The visitor is not authenticated and does not already have an account associated with the provided email address.
Post-condition	A new candidate account is created with a pending status and a verification email is sent to the provided address.
Scenario Description	The visitor navigates to the registration page, fills in the required information, and submits the form. The system creates a pending account and sends an activation email. The visitor clicks the activation link to complete the registration.
Main Scenario (Success)	1. The visitor navigates to the registration page. 2. The visitor fills in the form : first name, last name, email address, and password. 3. The visitor submits the registration form. 4. The system validates the input data and checks that the email is not already registered. 5. The system creates a new candidate account with a pending activation status. 6. The system sends a verification email containing an activation link. 7. The visitor clicks the activation link from their email inbox. 8. The system activates the account and confirms successful registration.
Alternative Scenarios	**A1 — Email already registered :** The system displays an error message indicating that an account with this email already exists and suggests signing in instead. **A2 — Invalid or incomplete data :** The system highlights the invalid fields and requests the visitor to correct them before resubmitting. **A3 — Activation link expired :** The system informs the user that the link has expired and offers to resend a new verification email.

TABLE 3.2 – Textual Description of Use Case “Register”

3.3.2.2 Use Case Description : “Sign In”

Use Case : Sign In	
Use Case	Sign In
Initiating Actor	Candidate, HR Officer (Administrator), Manager
Goal	Allow a registered and active user to log into the Sopra Recruit platform using their credentials in order to access their personal space.
Pre-condition	The user has a registered and activated account on the platform.
Post-condition	The user is authenticated and redirected to their role-specific dashboard.
Scenario Description	The user navigates to the login page, enters their email address and password, and submits the form. The system validates the credentials, generates a JWT token, and establishes the user session.
Main Scenario (Success)	1. The user navigates to the login page. 2. The user enters their email address and password. 3. The user submits the login form. 4. The system verifies the credentials and checks the account status. 5. The system generates a JWT token and establishes the user session. 6. The system redirects the user to their role-specific dashboard (Candidate, HR Officer, or Manager).
Alternative Scenarios	**A1 — Invalid credentials :** The system displays an error message and invites the user to try again. **A2 — Account not yet activated :** The system informs the user that their account has not been activated and offers to resend the verification email. **A3 — Account suspended :** The system displays a message indicating that the account has been suspended by the administrator.

TABLE 3.3 – Textual Description of Use Case “Sign In”

3.3.2.3 Use Case Description : “Manage User Account”

Use Case : Manage User Account	
Use Case	Manage User Account
Initiating Actor	HR Officer (Administrator)
Goal	Allow the HR administrator to create, view, update, and deactivate user accounts on the platform in order to control access and maintain an accurate user directory.
Pre-condition	The HR officer is authenticated and has administrative access to the user management interface.
Post-condition	The requested user account operation is successfully performed and the changes are persisted in the system.
Scenario Description	The HR officer accesses the user management interface, consults the list of existing accounts, and performs the desired operation : creating a new user, updating an existing account, or deactivating a user.
Main Scenario (Success)	1. The HR officer navigates to the user management interface. 2. The HR officer views the list of existing user accounts. 3. The HR officer selects an action : create a new user, update an existing account, or deactivate an account. 4. The HR officer fills in or modifies the required information (name, email, role). 5. The system validates the provided data. 6. The system saves the changes and displays a confirmation message.
Alternative Scenarios	**A1 — Duplicate email :** The system displays an error indicating that the email address is already associated with an existing account. **A2 — Invalid data :** The system highlights the invalid fields and requests the HR officer to correct them. **A3 — Deactivating the last active administrator :** The system prevents the operation and displays a warning to maintain platform integrity.

TABLE 3.4 – Textual Description of Use Case “Manage User Account”

3.3.2.4 Use Case Description : “Manage Permission”

Use Case : Manage Permission	
Use Case	Manage Permission
Initiating Actor	HR Officer (Administrator)
Goal	Allow the HR administrator to consult all platform roles and their associated permissions, and to control access by adding an existing permission to a role or disabling a permission from a role. The HR officer cannot create new permissions ; only pre-defined permissions can be managed.
Pre-condition	The HR officer is authenticated and has access to the permission management interface. The list of roles and the list of available permissions already exist in the system.
Post-condition	The selected role’s permission set is updated : the chosen permission is either active (added) or disabled (removed) for that role, and the change takes effect immediately across the platform.
Scenario Description	The HR officer accesses the permission management interface, which displays all roles (Candidate, HR Officer, Manager) alongside the permissions assigned to each. The HR officer selects a role and clicks “Modify” to open its permission list. They can then enable an existing permission by adding it to the role, or disable an existing permission to revoke access for that role.
Main Scenario (Success)	1. The HR officer navigates to the permission management interface. 2. The system displays all roles and their currently assigned permissions. 3. The HR officer selects a role and clicks “Modify”. 4. The system displays the full list of existing permissions, indicating which are currently enabled for that role. 5. The HR officer adds an existing permission to the role (enables it) or disables an existing permission (removes access). 6. The HR officer confirms the changes. 7. The system saves the updated permission set for the role and displays a confirmation message.
Alternative Scenarios	**A1 — No change made :** The HR officer closes the modification panel without applying any change ; the role’s permissions remain unchanged. **A2 — Disabling a critical permission :** The system warns the HR officer if disabling a permission may restrict core functionality for users with that role, and asks for confirmation before proceeding.

TABLE 3.5 – Textual Description of Use Case “Manage Permission”

3.4 Design

3.4.1 Sequence Diagrams

A sequence diagram is a UML behavioral diagram that shows how objects interact with each other over time, illustrating the order of messages exchanged between the system components for a given use case scenario [12].

3.4.1.1 Sequence Diagram : "Register"

Figure 3.2 presents the sequence diagram for the Register use case, showing the interactions between the visitor, the frontend interface, the backend server, and the email service during the account creation and activation process.

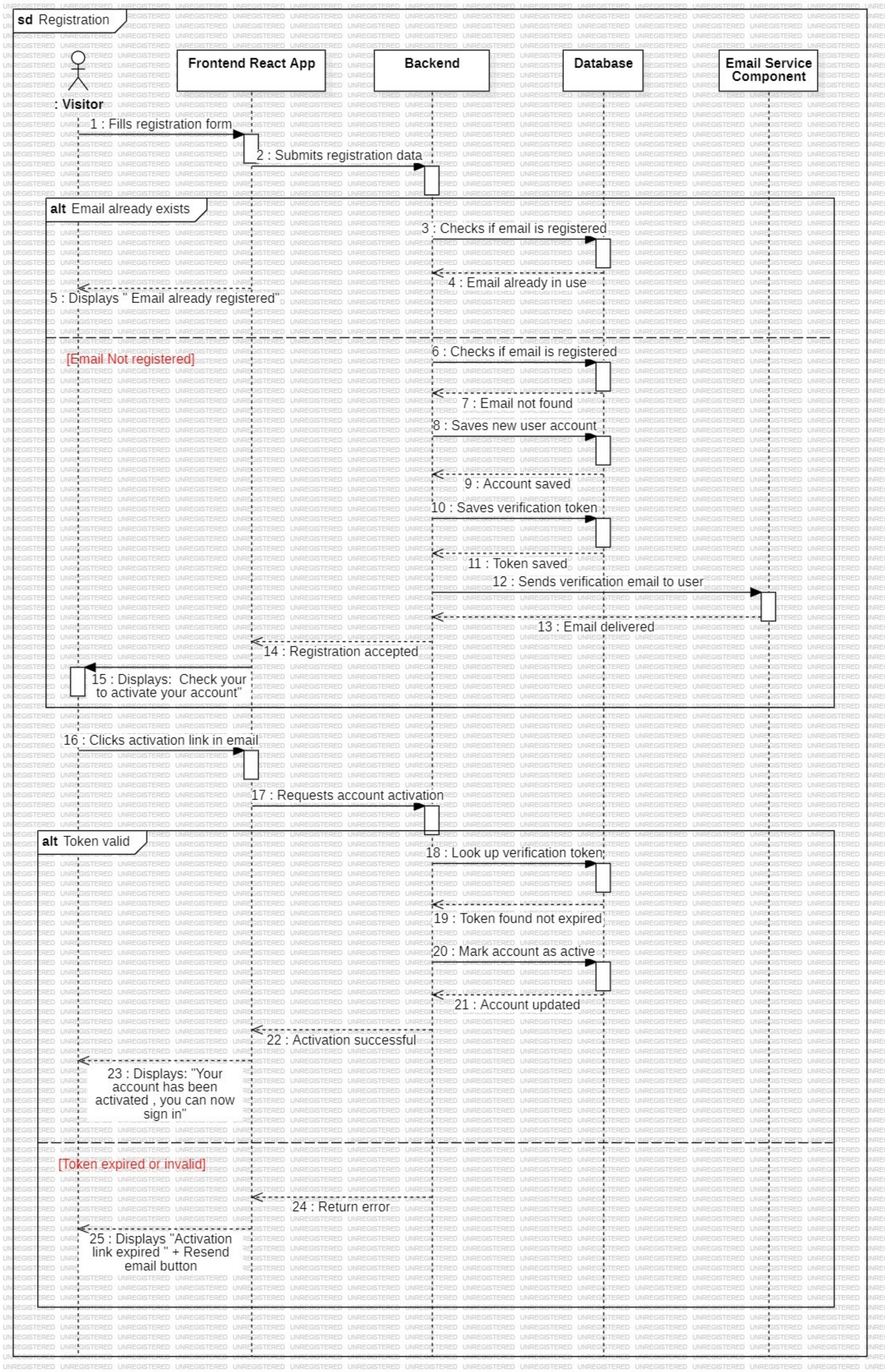

FIGURE 3.2 – Sequence Diagram — Register

3.4.1.2 Sequence Diagram : "Sign In"

Figure 3.3 presents the sequence diagram for the Sign In use case, illustrating the credential validation flow, JWT token generation, and role-based redirection performed between the client and the server.

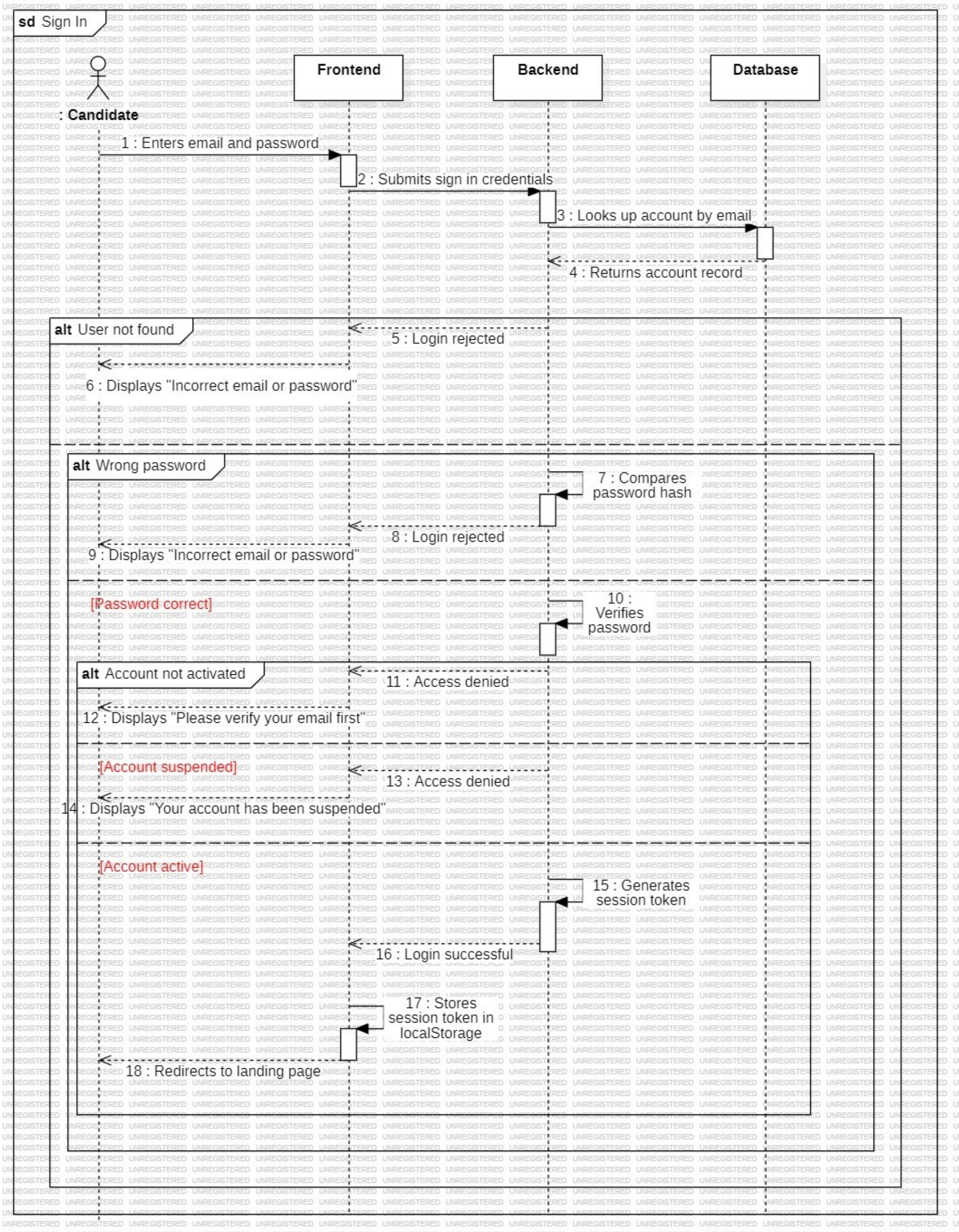

FIGURE 3.3 – Sequence Diagram — Sign In

3.4.1.3 Sequence Diagram : “Manage User Account”

Figure 3.4 presents the sequence diagram for the Manage User Account use case, detailing the interactions between the HR officer, the administration interface, and the backend during user creation, update, and deactivation operations.

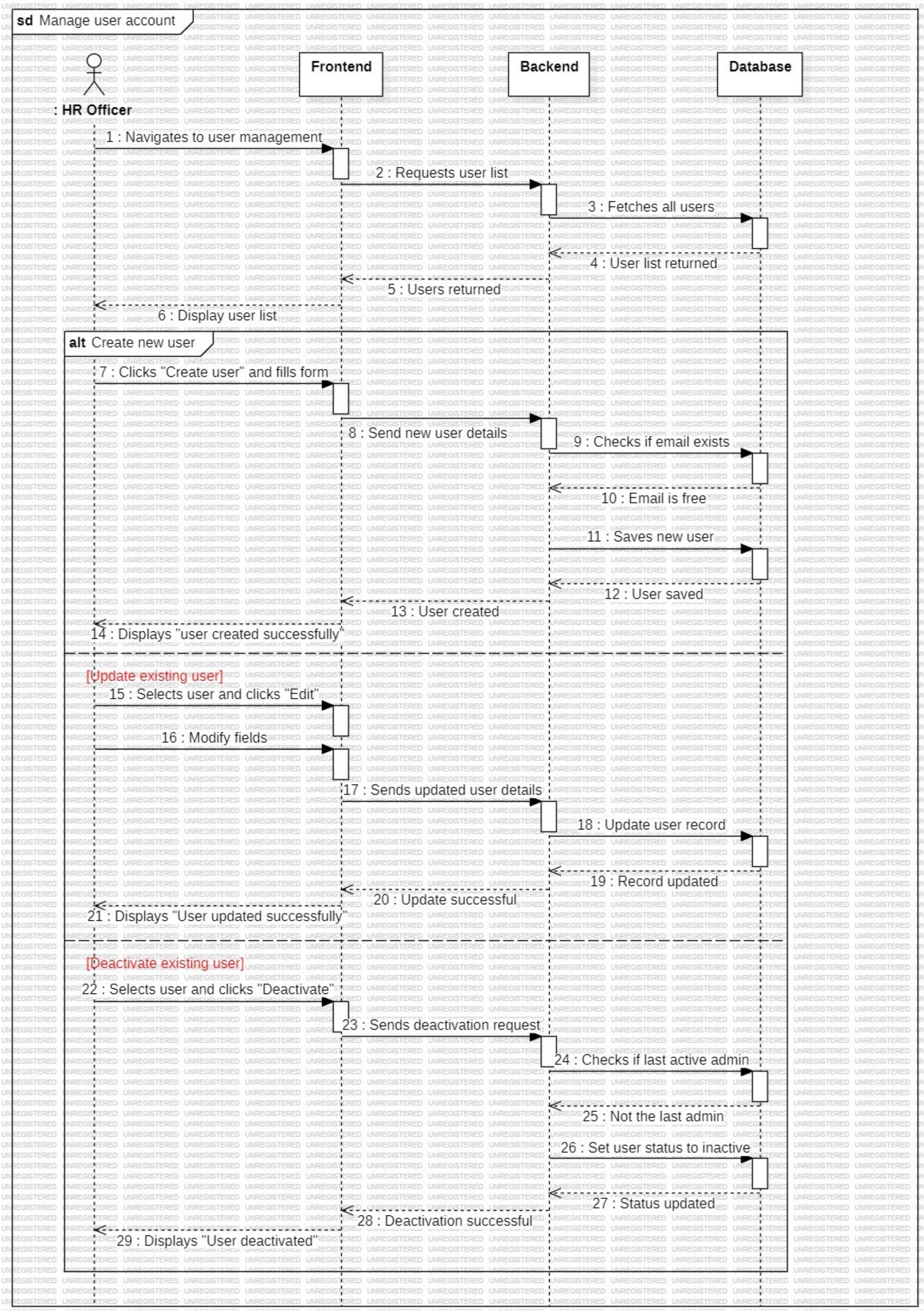

FIGURE 3.4 – Sequence Diagram — Manage User Account

3.4.1.4 Sequence Diagram : “Manage Permission”

Figure 3.5 presents the sequence diagram for the Manage Permission use case, showing the role assignment flow between the HR officer, the administration interface, and the backend permission management service.

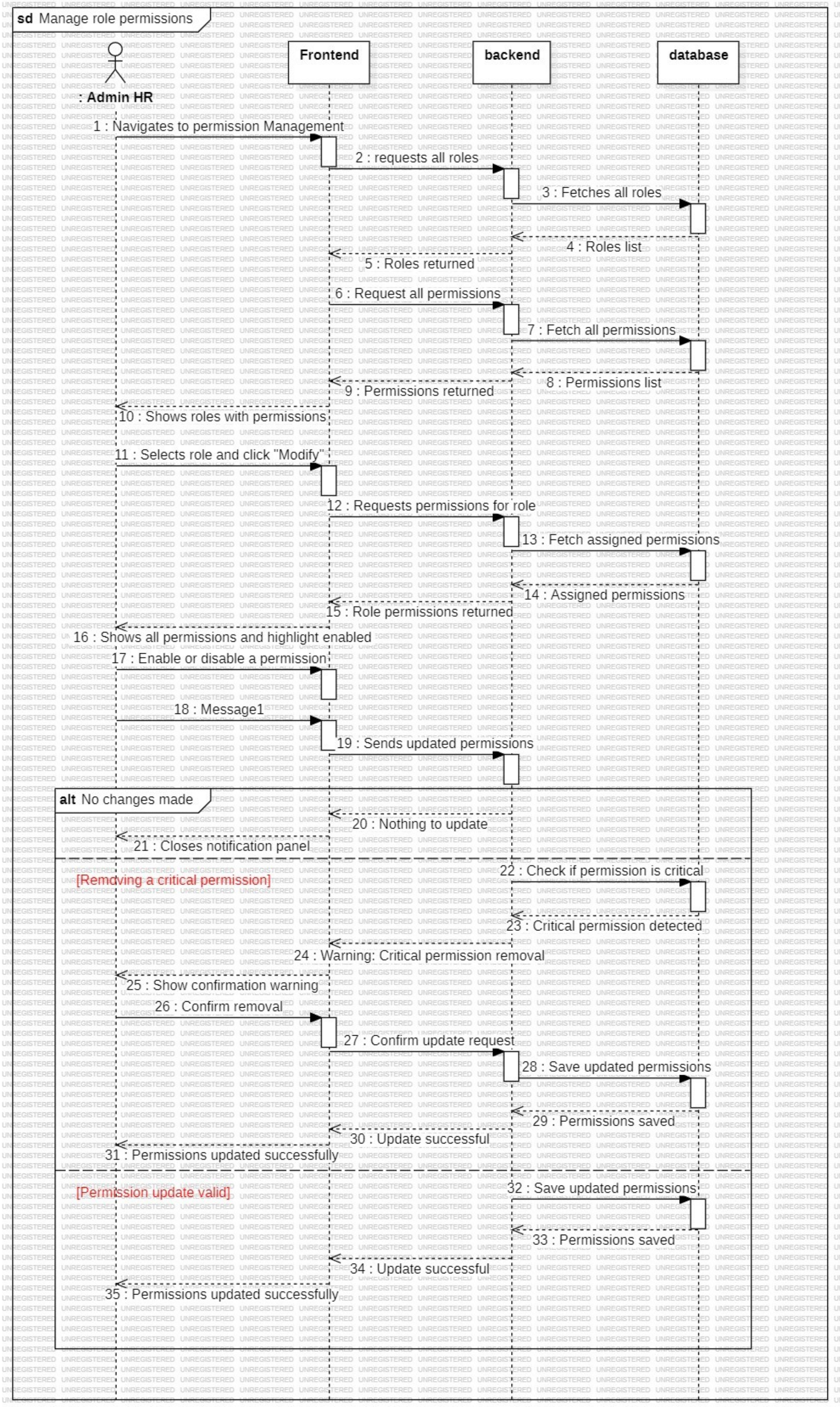

FIGURE 3.5 – Sequence Diagram — Manage Permission

3.4.2 Class Diagram

The class diagram below illustrates the conceptual structure of the entities involved in Sprint 1, showing the classes related to authentication and user management, along with their attributes, methods, and relationships [13].

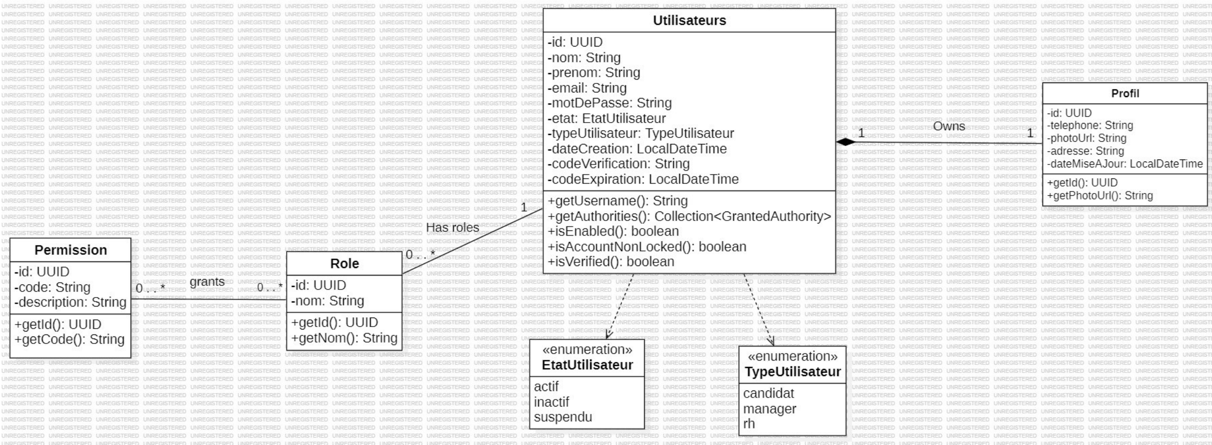

FIGURE 3.6 – Class Diagram — Sprint 1 : Authentication and User Management

3.5 Sprint Realization

This section presents the key user interfaces delivered at the end of Sprint 1. Each interface corresponds to a user story defined in the sprint backlog and illustrates the final visual result of the implemented features.

3.5.1 Registration Interface

The registration interface allows a visitor to create a new candidate account on the Sopra Recruit platform. The form collects the user's first name, last name, email address, and password. Upon submission, the system sends a verification email to activate the account. The interface includes inline validation to guide the user in providing correct information.

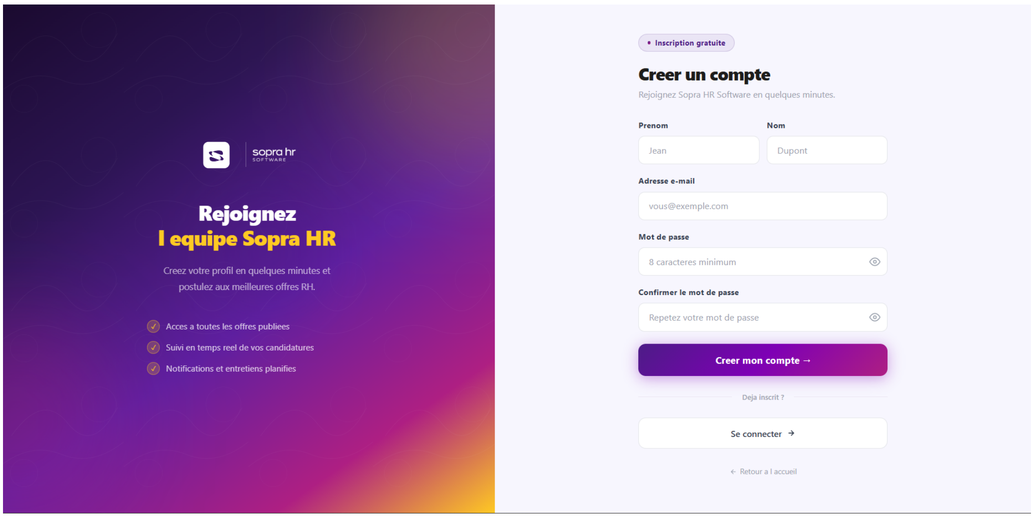

FIGURE 3.7 – Registration Interface

3.5.2 Email Verification Interface

After registration, the visitor receives a verification email containing an activation link. Once clicked, the system displays a confirmation page indicating that the account has been successfully activated and inviting the user to proceed to the sign-in page.

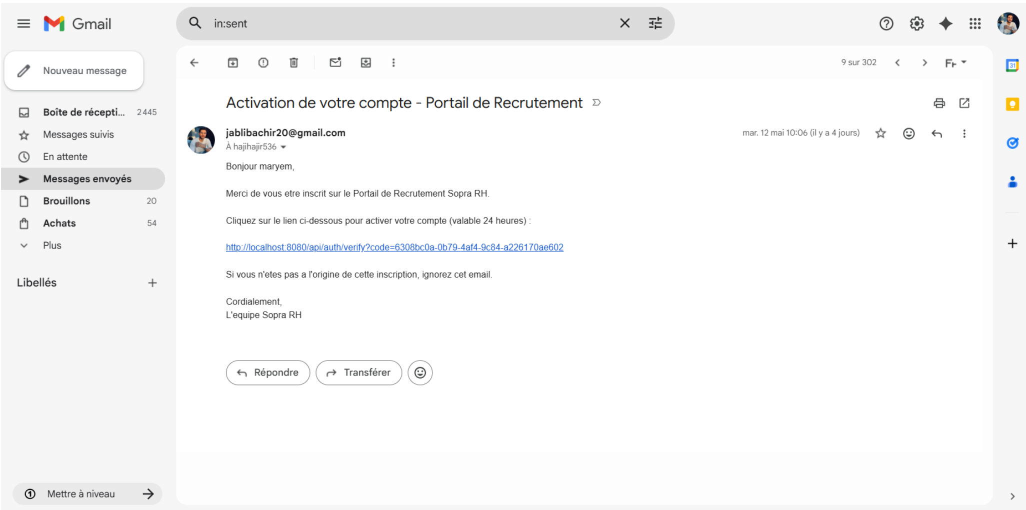

FIGURE 3.8 – Email Verification Confirmation Interface

3.5.3 Sign In Interface

The sign-in interface provides a secure login form for all platform roles — candidates, HR officers, and managers. The user enters their email address and password to authenticate. Upon successful login, the system generates a JWT token and redirects the user to their role-specific dashboard [9]. Error messages are displayed for invalid credentials or inactive accounts.

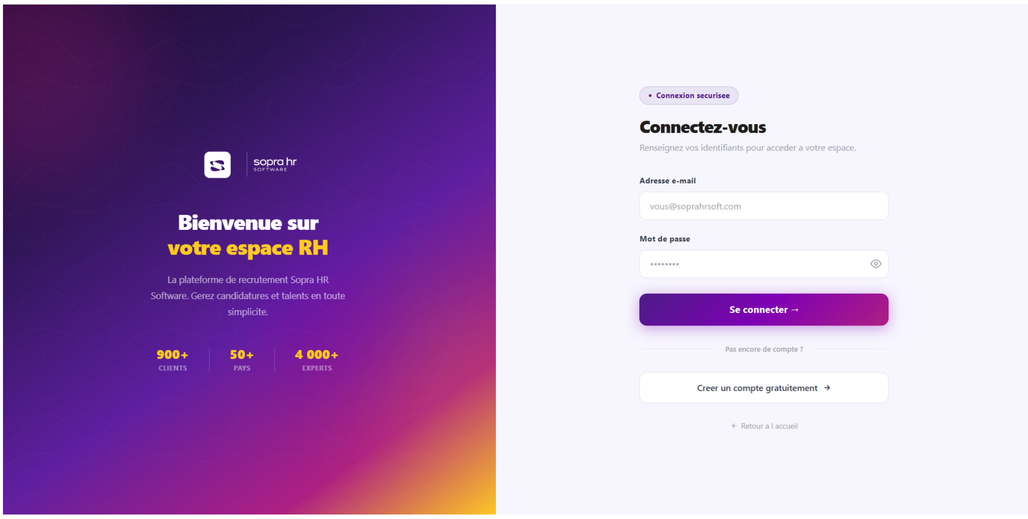

FIGURE 3.9 – Sign In Interface

3.5.4 User Management Interface

The user management interface is accessible exclusively to the HR officer. It displays the full list of platform users with their names, email addresses, roles, and account statuses. The HR officer can create new user accounts, edit existing ones, or deactivate accounts directly from this interface. A search bar is provided to quickly locate specific users.

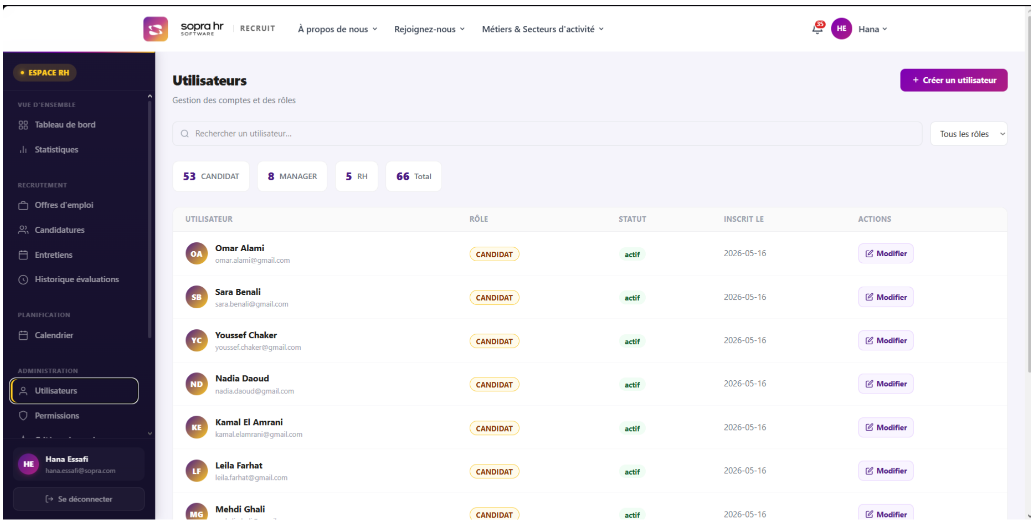

FIGURE 3.10 – User Management Interface

3.5.5 Role and Permission Management Interface

The role and permission management interface allows the HR officer to assign or modify the role of any platform user. A dropdown menu presents the available roles (Candidate, HR Officer, Manager), and the system enforces a safety rule that prevents removing the last active administrator. Changes are saved immediately and reflected across the platform.

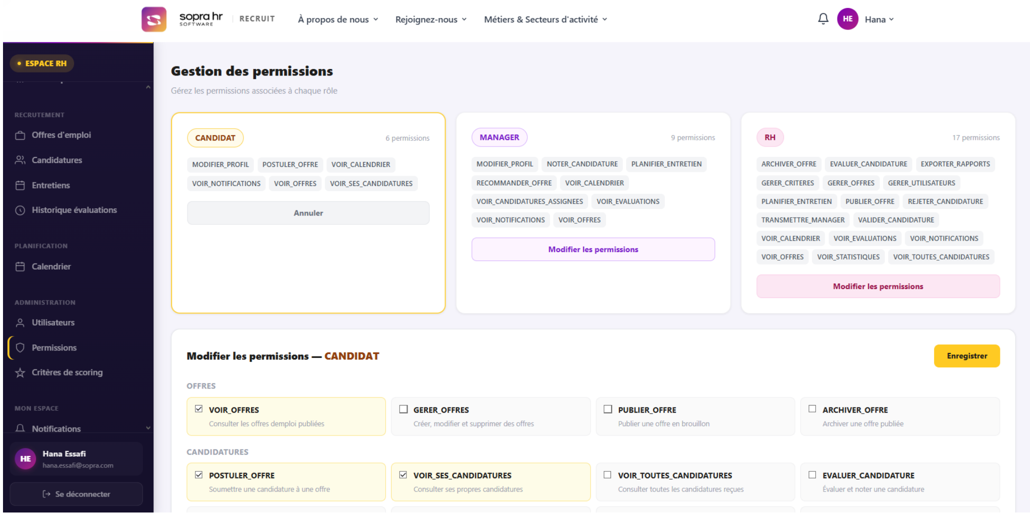

FIGURE 3.11 – Role and Permission Management Interface

3.5.6 Profile Management Interface

The profile management interface is available to all authenticated users. It allows them to view and update their personal information, including their phone number, address, and profile picture. Changes are saved in real time and immediately visible across the platform.

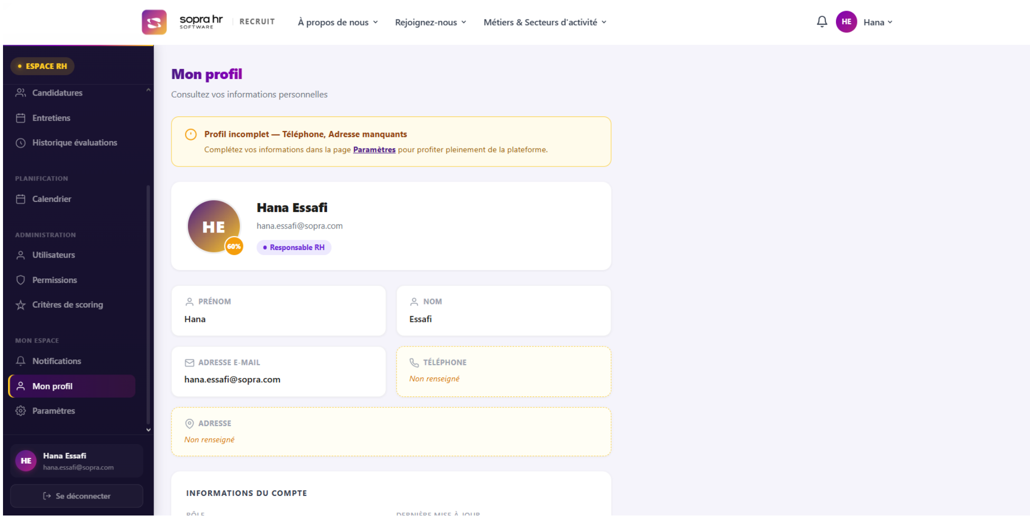

FIGURE 3.12 – Profile Management Interface

3.5.7 Landing Page

The landing page serves as the public entry point of the Sopra Recruit platform. It presents the company identity, highlights available job opportunities, and guides visitors toward registration or job browsing. The page features a clean and professional design consistent with the Sopra HR brand identity.

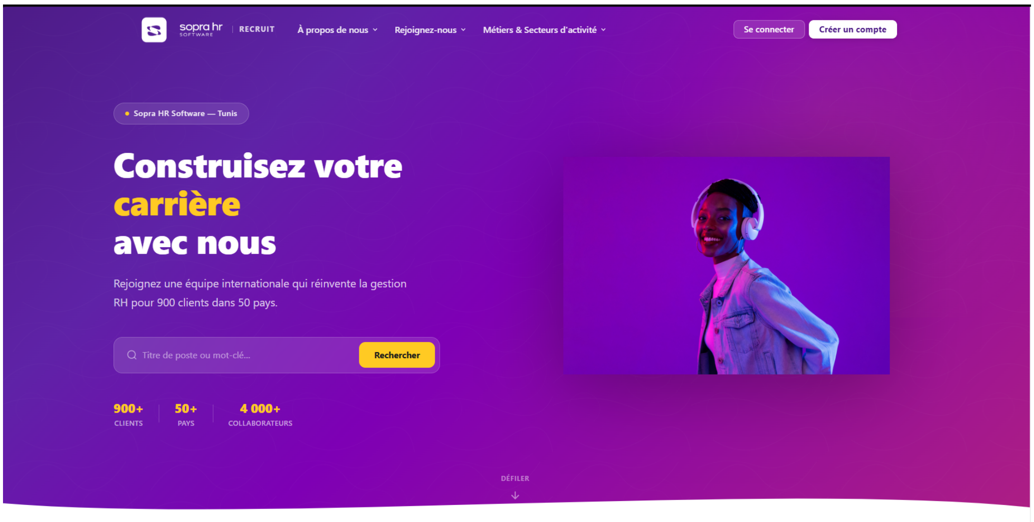

FIGURE 3.13 – Landing Page

Conclusion

This chapter presented the complete realization of Sprint 1 of the Sopra Recruit project. Starting from the sprint backlog and objectives, the analysis phase covered the use case diagram and detailed textual descriptions of the four main use cases : Register, Sign In, Manage User Account, and Manage Permission. The design phase introduced the sequence diagrams illustrating the interaction flows for each use case, as well as the class diagram representing the structural foundation of the authentication and user management modules. Finally, the sprint realization section showcased the implemented user interfaces, demonstrating that all planned user stories were successfully delivered. The platform's authentication layer and user management system are now fully operational and serve as the foundation for the development phases carried out in the subsequent sprints.

Chapter

4

SPRINT 2 : RECRUITMENT MANAGEMENT

Plan

Introduction

Sprint 2 covers the core recruitment management features of the Sopra Recruit platform. It encompasses the full lifecycle of job offers, from creation and publication to archiving, as well as candidate-facing browsing and filtering of available positions. This sprint also implements the application submission workflow, allowing candidates to upload their CV and cover letter, and introduces the application pipeline management tools available to HR officers and managers. Finally, the manager offer recommendation mechanism is delivered, enabling internal recruitment needs to be formally communicated and validated by HR. Together, these features form the operational heart of the recruitment platform and build directly on the authentication layer established in Sprint 1.

4.1 Sprint Backlog

Table 4.1 presents the sprint backlog for Sprint 2, detailing the user stories, their estimated duration, the associated implementation tasks, and the difficulty level assigned to each item.

User Story	Duration	Task	Difficulty
Job offer creation, publication and archiving	1 week	Build the offer creation form with title, description, required skills, and contract type ; implement status management (draft, published, archived) ; restrict published offer visibility to candidates.	High
Offer browsing with filters	4 days	Develop the public offer listing page ; implement keyword, experience level, and contract type filters ; display offer detail view accessible without authentication.	Medium
Application submission with CV and cover letter	1 week	Implement the multi-step application form ; handle CV upload and storage ; allow reuse of an existing CV ; record application date and initial status.	High
Application pipeline management	1 week	Build the HR application list with status, score, and date filters ; implement status transitions (under evaluation, forwarded to manager, rejected) ; allow CV download ; build the manager review interface for forwarded applications.	High
Manager offer recommendations	3 days	Develop the recommendation submission form for managers ; implement the HR review interface to accept or reject recommendations ; trigger automatic offer creation upon acceptance.	Medium

TABLE 4.1 – Sprint 2 Backlog

4.2 Sprint Objective

The primary objective of Sprint 2 is to deliver a fully operational recruitment management system that enables the complete lifecycle of job offers and candidate applications within the Sopra Recruit platform. By the end of this sprint, HR officers will be able to create, publish, and archive job offers, and will have access to a comprehensive application pipeline with status management and CV download capabilities. Visitors and candidates will be able to browse and filter published positions, and candidates will be able to submit applications with their CV and cover letter in a guided, multi-step process. Managers will be able to review applications forwarded by HR and either validate or reject them, as well as submit job offer recommendations that HR can approve or decline. These features transform the platform into a functional end-to-end recruitment tool and lay the groundwork for the AI-powered analysis features of Sprint 3.

4.3 Analysis

4.3.1 Use Case Diagram

A use case diagram is a UML diagram that illustrates the interactions between users (called actors) and the system. It describes the different functionalities (use cases) that the system offers to its users, showing who uses what and how, without going into technical details [11].

The use case diagram for Sprint 2, presented in Figure 4.1, identifies the interactions between the four system actors — Visitor, Candidate, HR Officer, and Manager — and the recruitment management features delivered during this sprint.

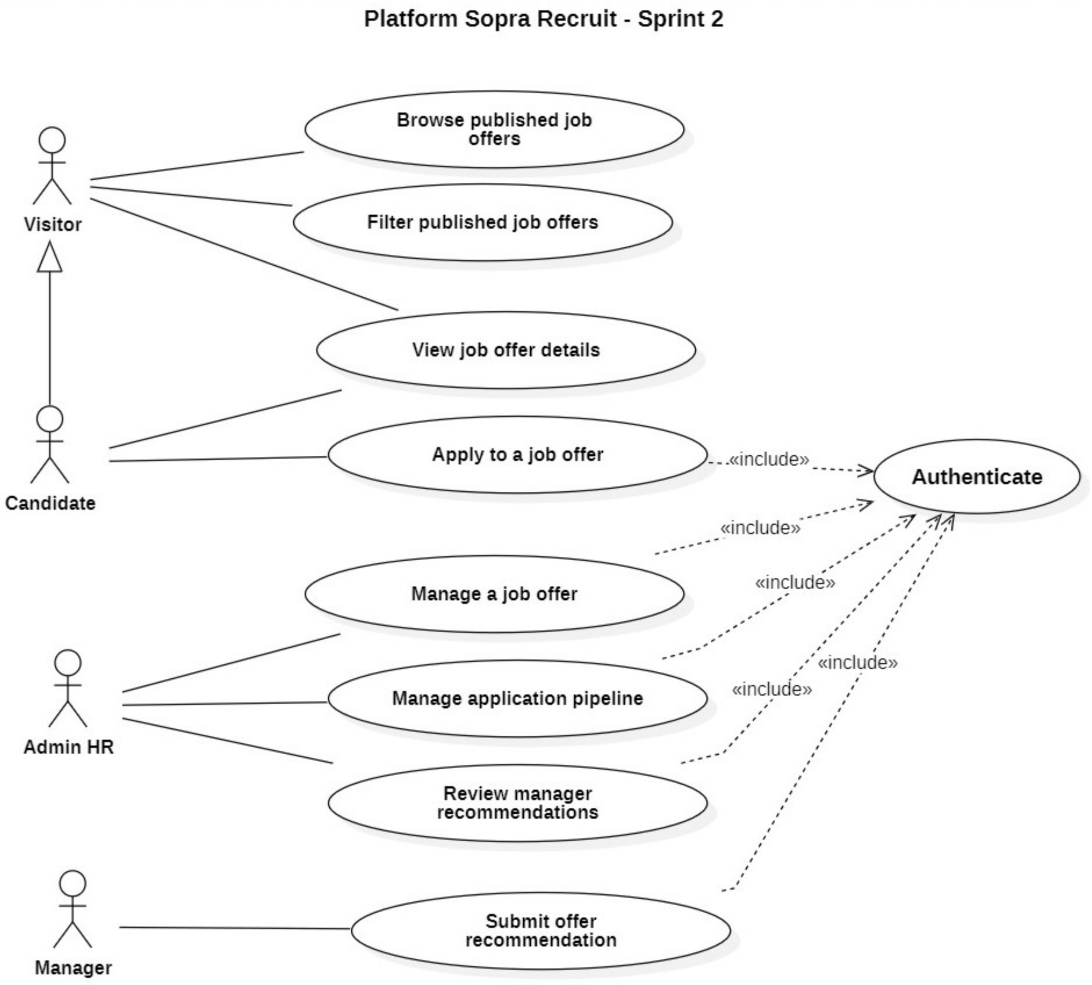

FIGURE 4.1 – Use Case Diagram — Sprint 2 : Recruitment Management

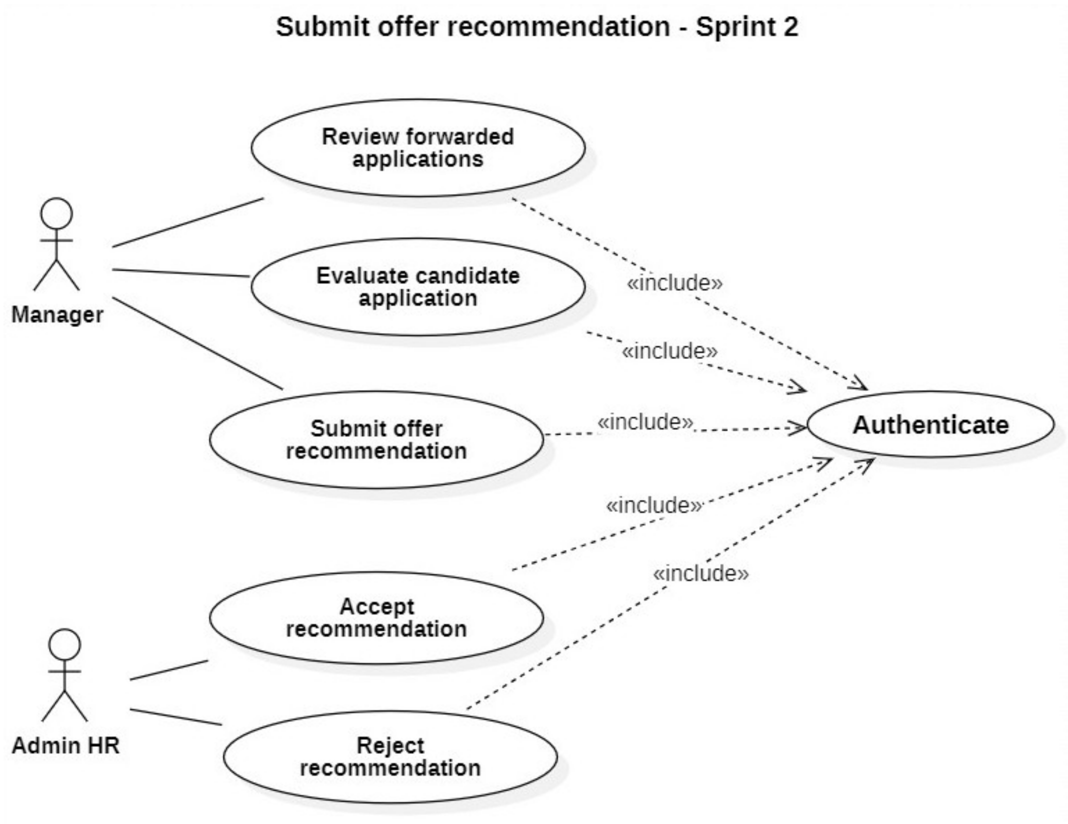

FIGURE 4.2 – Use Case Diagram — Apply to Job Offer (Refinement)

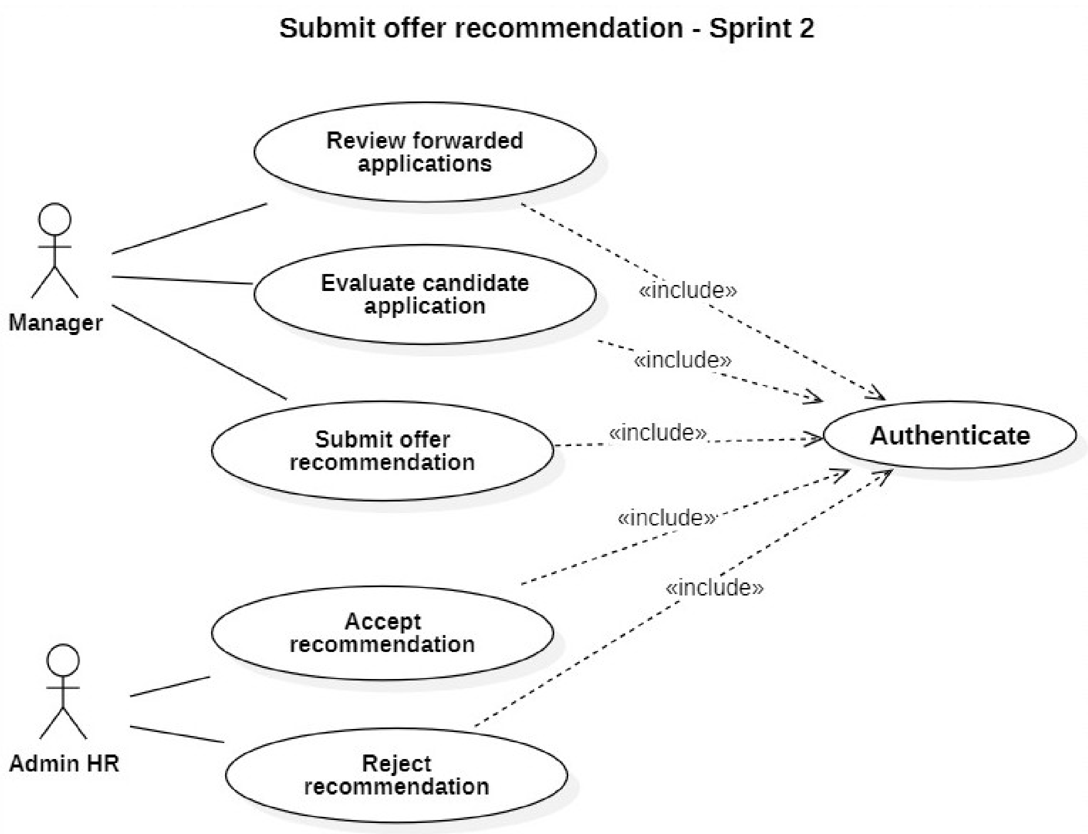

FIGURE 4.3 – Use Case Diagram — Submit Offer Recommendation (Refinement)

4.3.2 Textual Description of Use Cases

This section provides a textual description of the main use cases appearing in the Sprint 2 use case diagram. Each use case is described through its initiating actor, goal, pre-conditions, post-conditions, main success scenario, and alternative scenarios.

4.3.2.1 Use Case Description : "Manage Job Offer"

Use Case : Manage Job Offer	
Use Case	Manage Job Offer
Initiating Actor	HR Officer (Administrator)
Goal	Allow the HR officer to create, update, publish, and archive job offers so that available positions are communicated accurately to candidates on the platform.
Pre-condition	The HR officer is authenticated and has access to the offer management interface.
Post-condition	The job offer is created or updated with the desired status and, if published, becomes visible to visitors and candidates on the public listing page.
Scenario Description	The HR officer navigates to the offer management interface, fills in the offer details, and selects the appropriate status. Published offers become immediately visible to unauthenticated visitors and candidates.
Main Scenario (Success)	1. The HR officer navigates to the job offer management interface. 2. The HR officer clicks "Create New Offer". 3. The HR officer fills in the offer form : title, description, required skills, experience level, contract type, and location. 4. The HR officer selects the offer status : draft, published, or archived. 5. The HR officer submits the form. 6. The system validates the data and saves the offer. 7. If the status is "published", the offer becomes visible on the public listing page immediately. 8. The system displays a confirmation message and redirects to the offer list.
Alternative Scenarios	**A1 — Incomplete form :** The system highlights the missing required fields and asks the HR officer to complete them before saving. **A2 — Archive a published offer :** The HR officer changes the status to "archived" ; the offer is immediately removed from the public listing. **A3 — Edit an existing offer :** The HR officer selects an existing offer, modifies the desired fields, and saves the changes.

TABLE 4.2 – Textual Description of Use Case "Manage Job Offer"

4.3.2.2 Use Case Description : "Apply to Job Offer"

Use Case : Apply to Job Offer	
Use Case	Apply to Job Offer
Initiating Actor	Candidate
Goal	Allow an authenticated candidate to submit an application for a published job offer by providing a CV and optional cover letter, so that their profile enters the recruitment pipeline.
Pre-condition	The candidate is authenticated. The targeted job offer exists and is published. The candidate has not already applied to this offer.
Post-condition	A new application is recorded in the system with the status "Under Evaluation". The submitted CV is stored and linked to the application. The HR officer can view the new application in the pipeline.
Scenario Description	The candidate navigates to the offer detail page, clicks "Apply", and is guided through a multi-step form to upload or reuse a CV and add a cover letter. The system records the application upon submission.
Main Scenario (Success)	1. The candidate navigates to the published offer detail page. 2. The candidate clicks the "Apply" button. 3. The system presents the application form (Step 1) : the candidate chooses to upload a new CV or reuse an existing one. 4. The candidate uploads the CV file (PDF format). 5. The system presents the application form (Step 2) : the candidate optionally adds a cover letter. 6. The candidate reviews the application summary and confirms submission. 7. The system records the application with the status "Under Evaluation" and the submission date. 8. The system confirms successful submission and redirects the candidate to their applications list.
Alternative Scenarios	**A1 — Candidate already applied :** The system detects the duplicate application and informs the candidate that they have already applied to this offer. **A2 — No CV provided :** The system prevents submission and displays a message requiring the candidate to provide a CV before proceeding. **A3 — Unsupported file format :** The system rejects the uploaded file and asks the candidate to upload a valid PDF document.

TABLE 4.3 – Textual Description of Use Case "Apply to Job Offer"

4.3.2.3 Use Case Description : "Manage Application Pipeline"

Use Case : Manage Application Pipeline	
Use Case	Manage Application Pipeline
Initiating Actor	HR Officer (Administrator)
Goal	Allow the HR officer to view, filter, and update the status of all received applications so that qualified candidates are progressed through the recruitment workflow and unqualified ones are rejected.
Pre-condition	The HR officer is authenticated. At least one application exists in the system.
Post-condition	The application status is updated. If forwarded to a manager, the application becomes visible in the manager's review interface. If rejected, the candidate's application status is updated accordingly.
Scenario Description	The HR officer opens the application management interface, filters and reviews applications, and transitions them through the pipeline by changing their status.
Main Scenario (Success)	1. The HR officer navigates to the application management interface. 2. The system displays all received applications with their status, submission date, and offer title. 3. The HR officer applies filters (by status, score, or date) to narrow the list. 4. The HR officer opens an application detail to review the candidate profile and CV. 5. The HR officer changes the application status : "Forward to Manager", or "Reject". 6. The system saves the updated status and records the transition date. 7. If forwarded, the application appears in the designated manager's review queue.
Alternative Scenarios	**A1 — Download CV :** The HR officer clicks "Download CV" to retrieve the candidate's original document for offline review. **A2 — No applications match filters :** The system displays an empty state message and suggests broadening the filter criteria.

TABLE 4.4 – Textual Description of Use Case "Manage Application Pipeline"

4.3.2.4 Use Case Description : "Submit Offer Recommendation"

Use Case : Submit Offer Recommendation	
Use Case	Submit Offer Recommendation
Initiating Actor	Manager
Goal	Allow a manager to formally propose a new job opening to the HR officer, so that recruitment needs identified at the department level are officially communicated and can be processed.
Pre-condition	The manager is authenticated and has access to the recommendation submission form.
Post-condition	A recommendation request is created and sent to the HR officer's review queue. The HR officer can then accept or reject the proposal.
Scenario Description	The manager fills in a recommendation form describing the position needed in their department, submits it to HR, and waits for a decision. The HR officer reviews the request and either accepts it, triggering offer creation, or rejects it with a justification.
Main Scenario (Success)	1. The manager navigates to the recommendation submission interface. 2. The manager fills in the form : proposed job title, department, required skills, and justification. 3. The manager submits the recommendation request. 4. The system records the request with a "Pending" status and notifies the HR officer. 5. The HR officer reviews the recommendation and clicks "Accept" or "Reject". 6. If accepted, the system automatically creates a draft job offer pre-filled with the recommendation data. 7. If rejected, the HR officer provides a rejection reason visible to the manager. 8. The system updates the recommendation status and notifies the manager of the decision.
Alternative Scenarios	**A1 — Incomplete recommendation form :** The system highlights required fields and prevents submission until all mandatory information is provided. **A2 — Duplicate recommendation :** The system warns the manager that a similar request for the same position is already pending and asks them to confirm before creating another one.

TABLE 4.5 – Textual Description of Use Case "Submit Offer Recommendation"

4.4 Design

4.4.1 Sequence Diagrams

A sequence diagram is a UML behavioral diagram that shows how objects interact with each other over time, illustrating the order of messages exchanged between the system components for a given use case scenario [12].

4.4.1.1 Sequence Diagram : "Manage Job Offer"

Figure 4.4 presents the sequence diagram for the Manage Job Offer use case, showing the interactions between the HR officer, the frontend interface, and the backend server during job offer creation and status management.

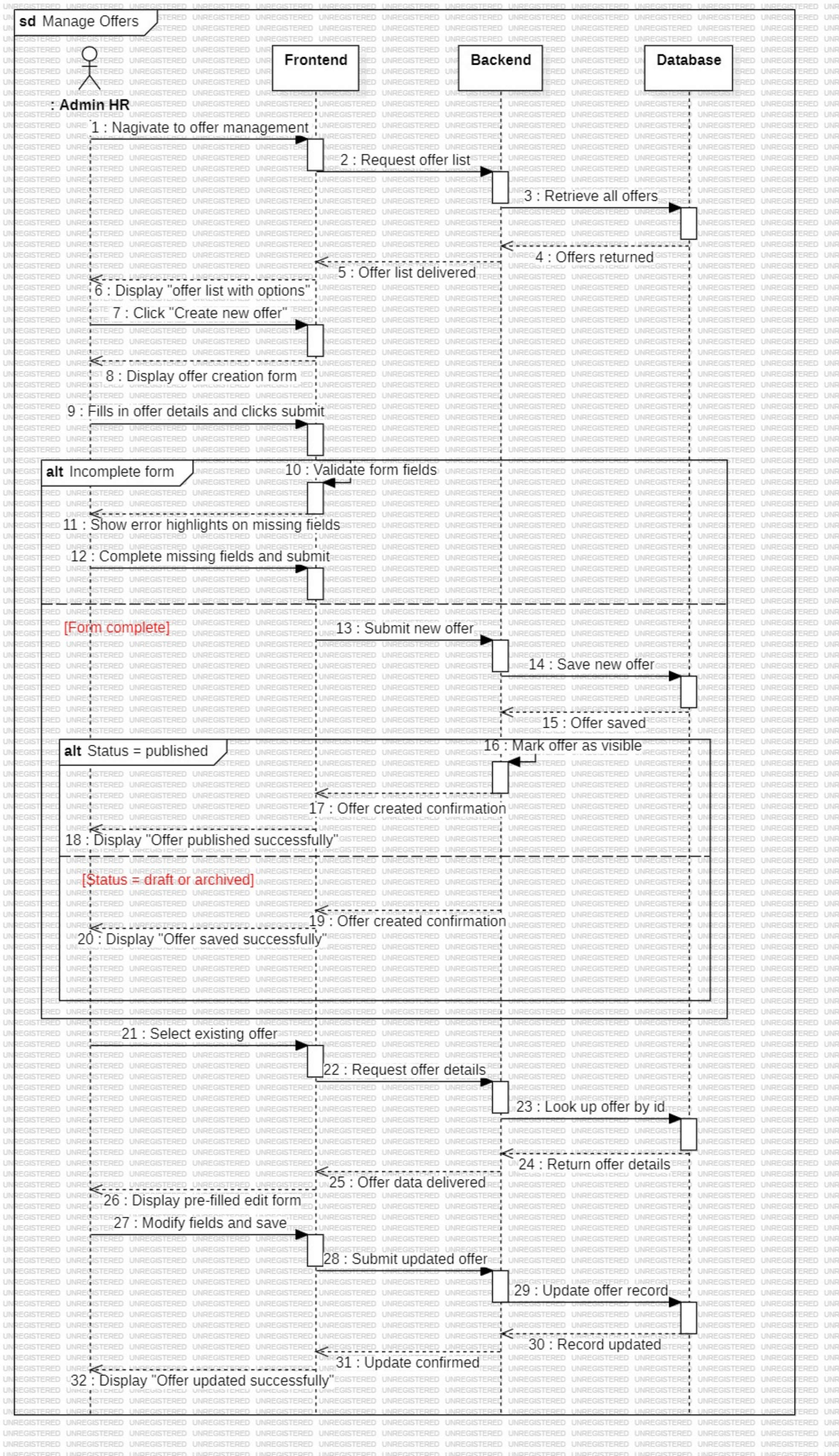

FIGURE 4.4 – Sequence Diagram — Manage Job Offer

4.4.1.2 Sequence Diagram : "Apply to Job Offer"

Figure 4.5 presents the sequence diagram for the Apply to Job Offer use case, illustrating the candidate application flow, CV upload handling, and confirmation process between the client, frontend, and backend.

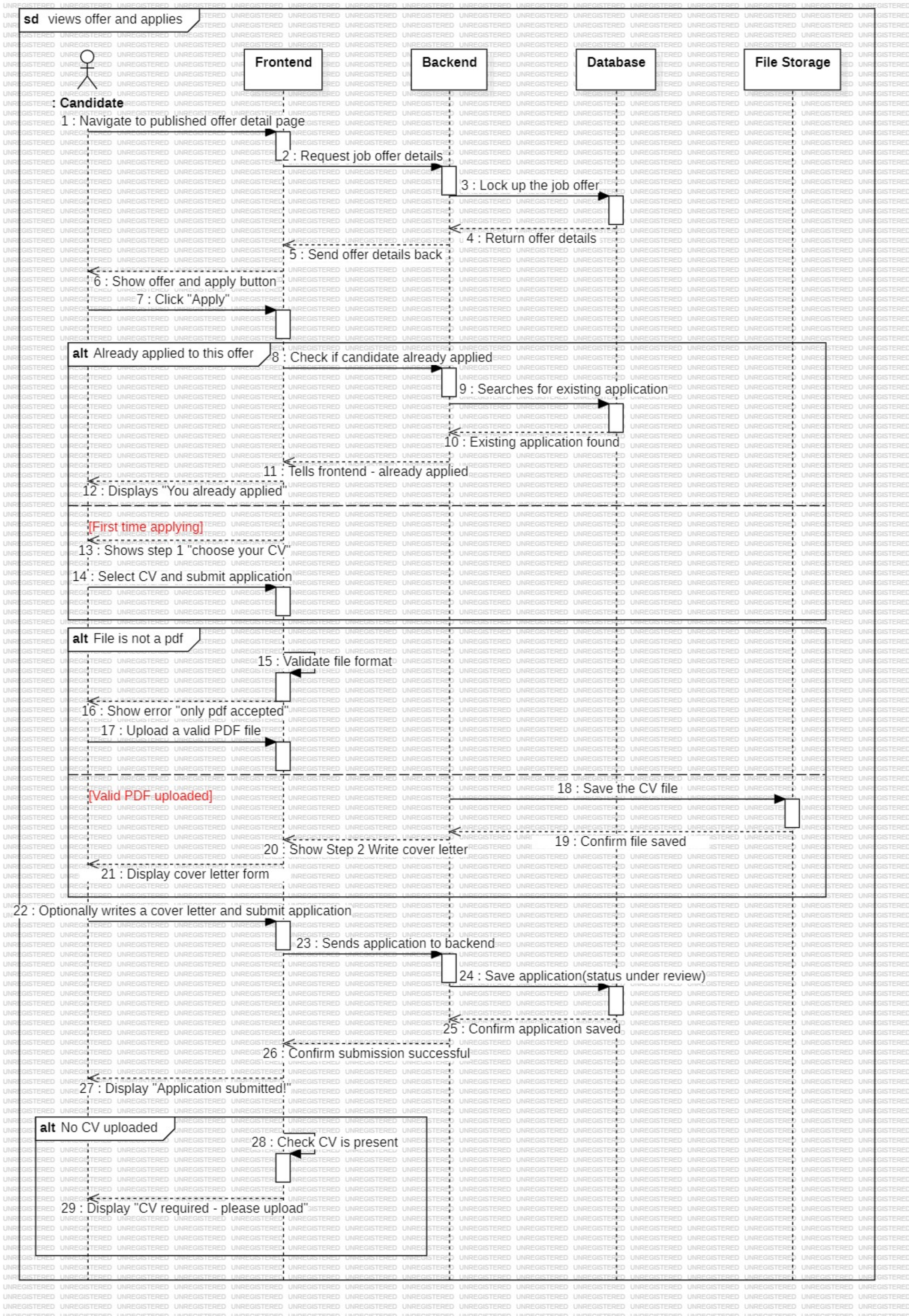

FIGURE 4.5 – Sequence Diagram — Apply to Job Offer

4.4.1.3 Sequence Diagram : "Manage Application Pipeline"

Figure 4.6 presents the sequence diagram for the Manage Application Pipeline use case, detailing the status transition flow between the HR officer, the application management interface, and the backend during pipeline operations.

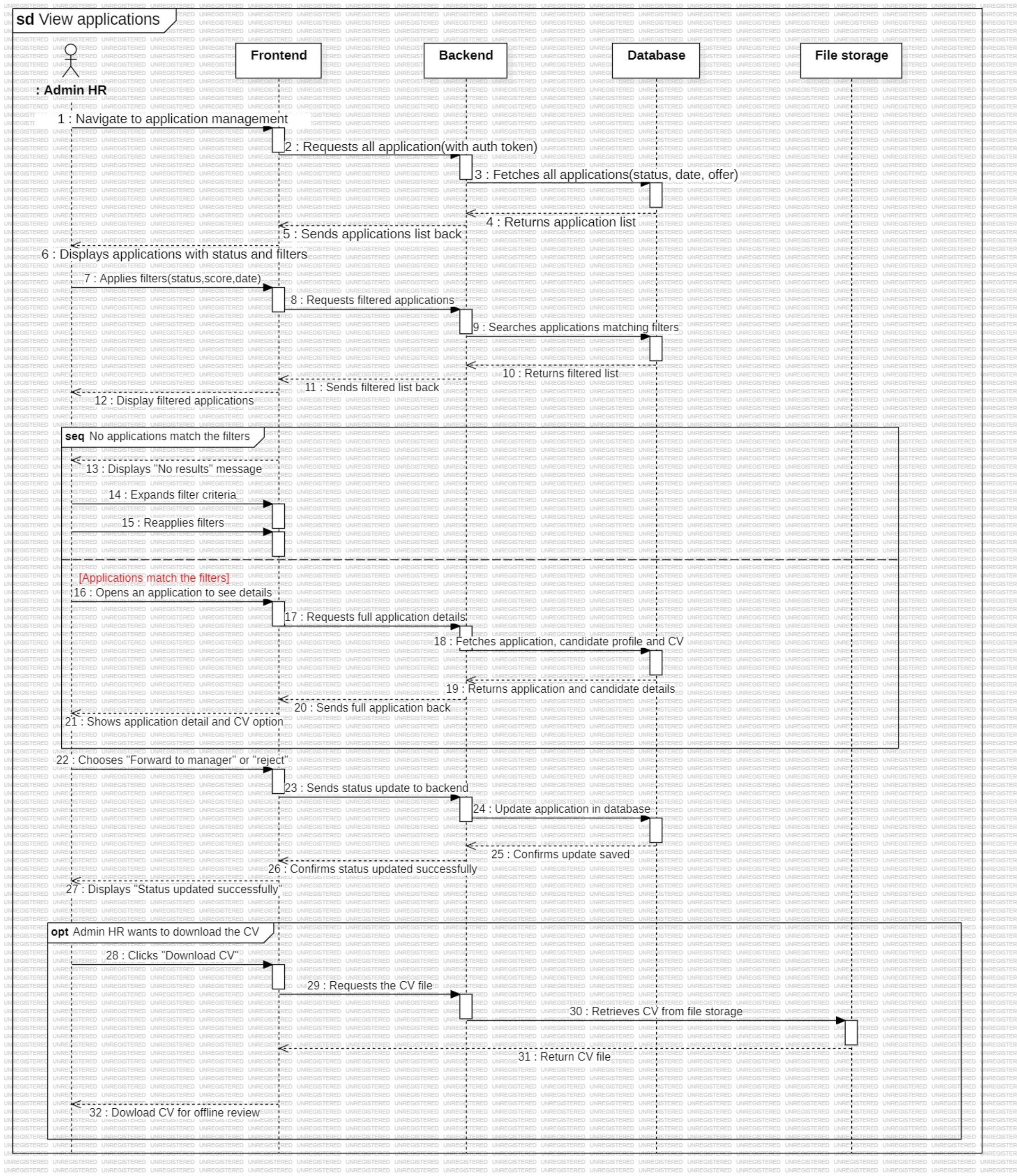

FIGURE 4.6 – Sequence Diagram — Manage Application Pipeline

4.4.1.4 Sequence Diagram : “Submit Offer Recommendation”

Figure 4.7 presents the sequence diagram for the Submit Offer Recommendation use case, showing the communication flow between the manager, the recommendation interface, the backend, and the HR officer during the proposal and review process.

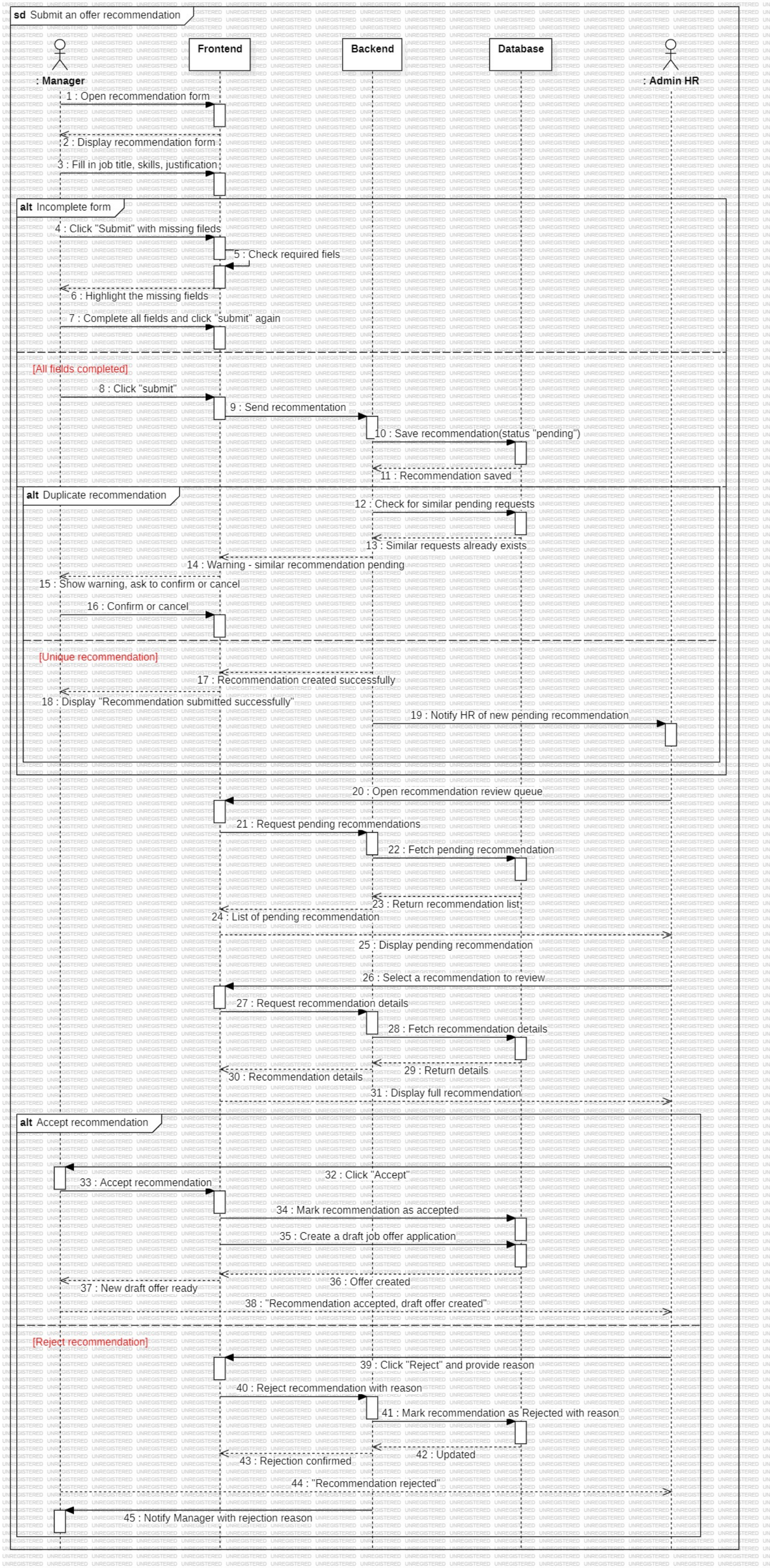

FIGURE 4.7 – Sequence Diagram — Submit Offer Recommendation

4.4.2 Class Diagram

The class diagram below illustrates the conceptual structure of the entities involved in Sprint 2, showing the classes related to job offers, applications, and recommendations, along with their attributes, methods, and relationships [13].

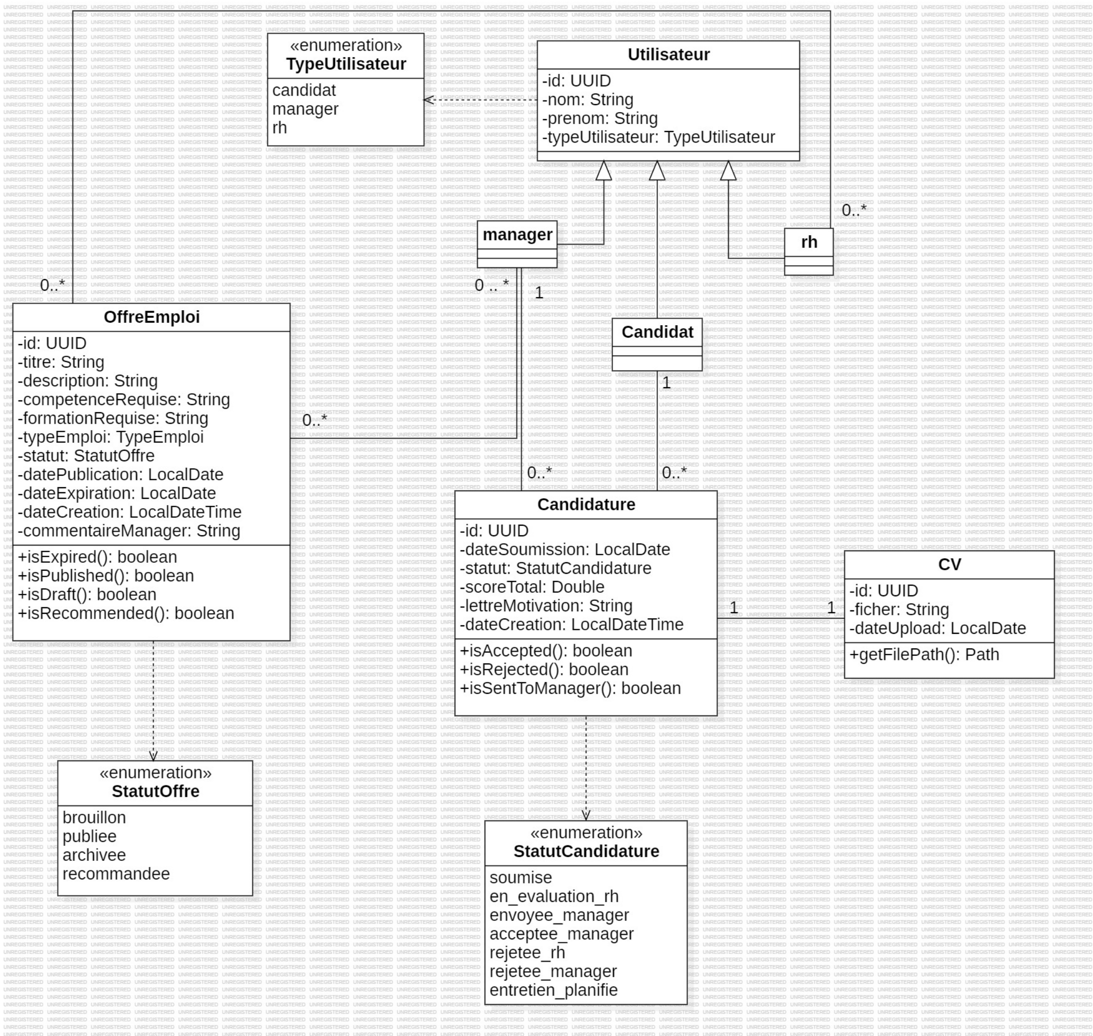

FIGURE 4.8 – Class Diagram — Sprint 2 : Recruitment Management

4.5 Sprint Realization

This section presents the key user interfaces delivered at the end of Sprint 2. Each interface corresponds to a user story defined in the sprint backlog and illustrates the final visual result of the implemented features.

4.5.1 Job Offer Management Interface (HR View)

The job offer management interface is accessible exclusively to the HR officer. It combines offer creation and offers listing in a single view : the officer can create a new offer by entering the title, description, required skills, experience level, contract type, and location, while also keeping a complete view of all existing offers. The list displays key information such as title, contract type, status, and creation date, and the officer can change the status of any offer or open its detail page directly from the same interface.

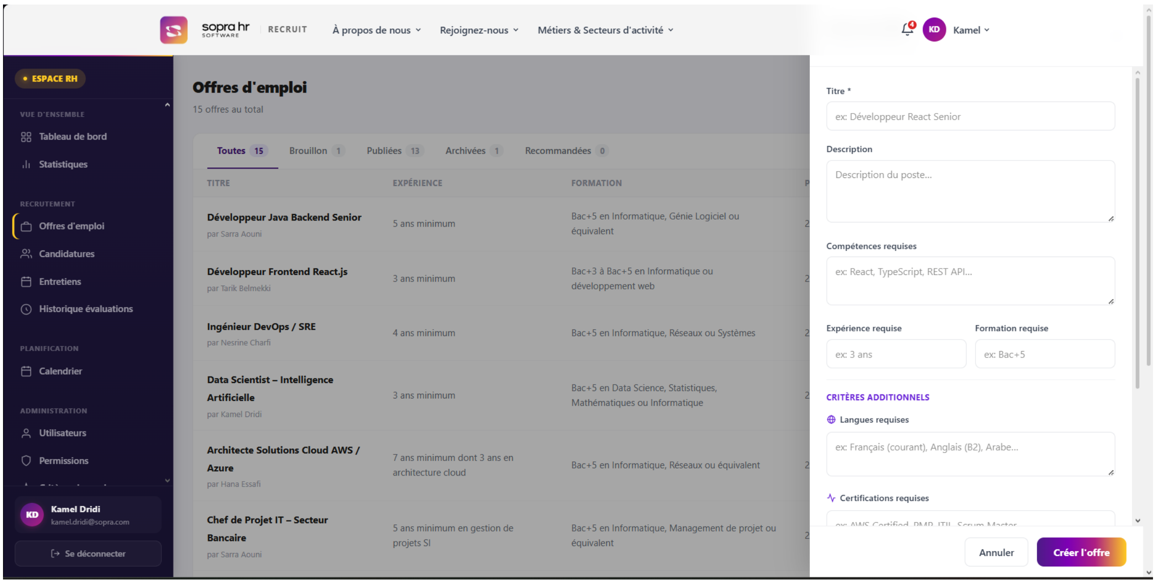

FIGURE 4.9 – Job Offer Management Interface — HR View

4.5.2 Public Job Offers Listing Interface

The public job offers listing interface is accessible to all visitors without authentication. It displays all currently published offers and provides filter controls for keyword search, experience level, and contract type. Each offer card shows a summary of the position, and clicking on it opens the full offer detail page.

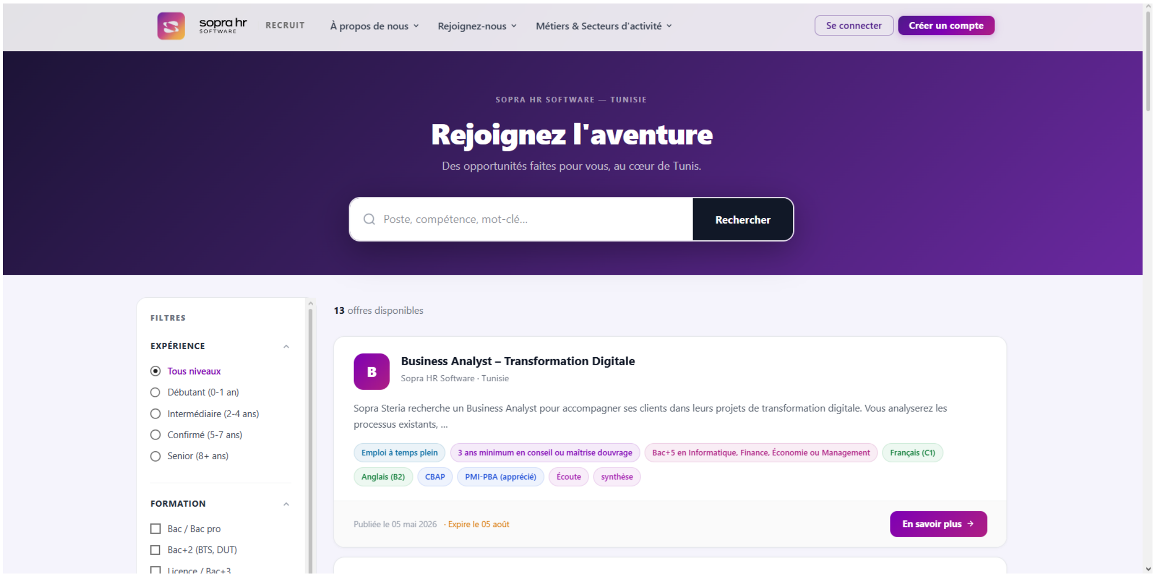

FIGURE 4.10 – Public Job Offers Listing Interface

4.5.3 Application Submission Interface

The application submission interface guides the candidate through a multi-step process to apply for a job offer. In the first step, the candidate uploads a new CV or selects a previously used one. In the second step, the candidate optionally adds a cover letter. A final summary step allows the candidate to review and confirm the application before it is submitted to the system.

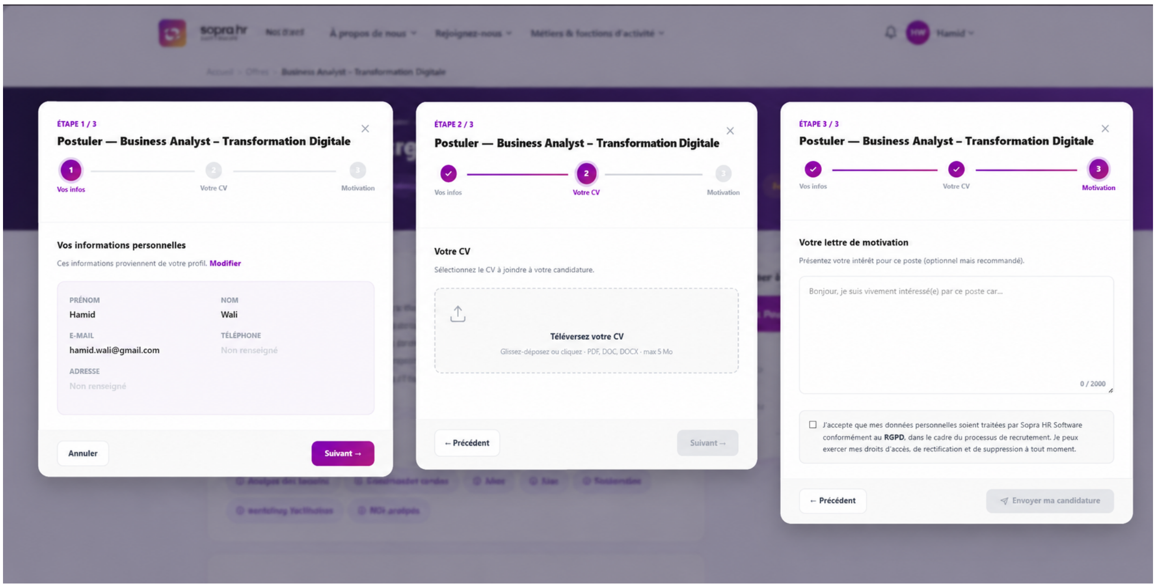

FIGURE 4.11 – Application Submission Interface

4.5.4 Candidate Applications List Interface

The candidate applications list interface allows authenticated candidates to track all of their submitted applications in one place. For each application, the interface displays the job title, submission date, and current status (e.g., Under Evaluation, Forwarded to Manager, Rejected). This gives candidates full visibility over their recruitment progress without needing to contact HR.

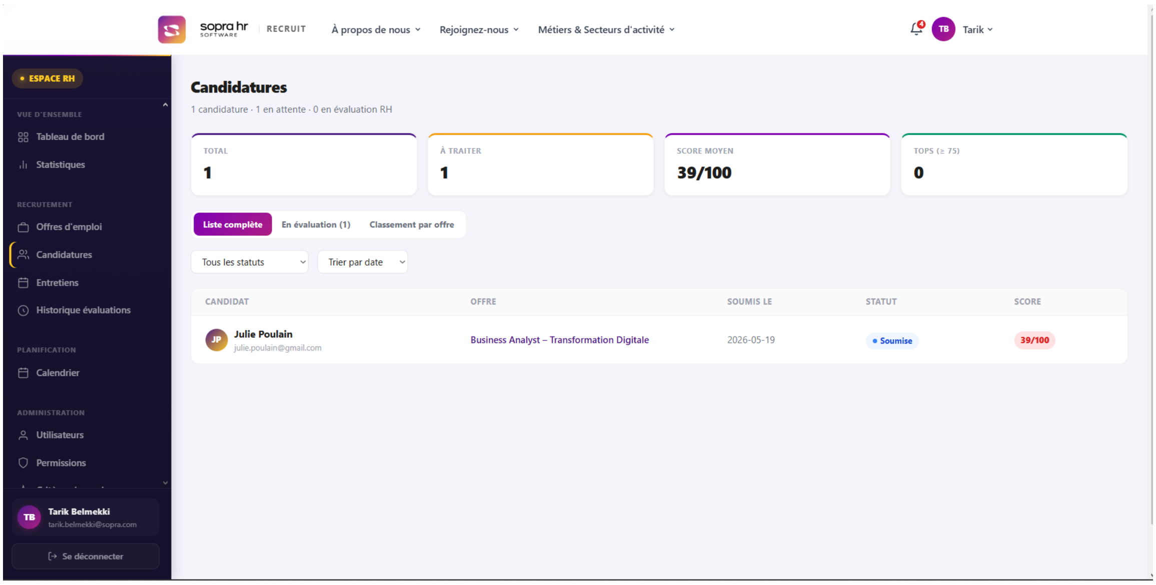

FIGURE 4.12 – Candidate Applications List Interface

4.5.5 HR Application Pipeline Interface

The HR application pipeline interface is the central management tool for the HR officer to oversee all received applications. It provides filtering by status, score, and date, displays candidate summaries, and allows the HR officer to change the status of any application : forwarding it to a manager or rejecting it. A CV download button is available for each application.

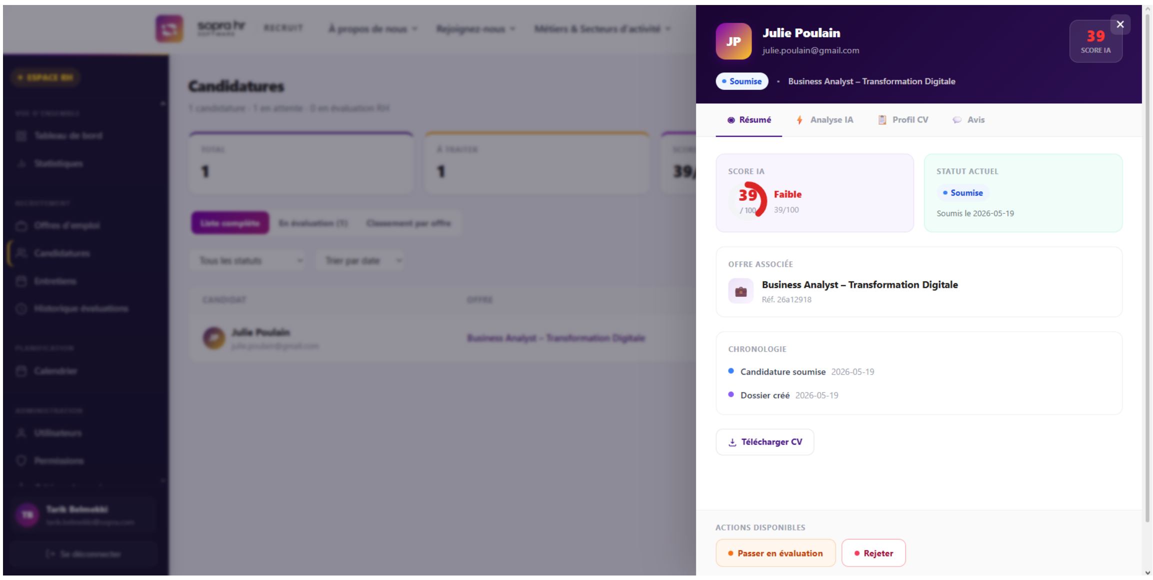

FIGURE 4.13 – HR Application Pipeline Interface

4.5.6 Offer Recommendation Interface

The offer recommendation interface allows managers to formally propose a new job opening to the HR department. The manager fills in a form describing the position, required skills, and the justification for the recruitment need. Once submitted, the request is sent to the HR officer, who can accept the proposal (generating a draft offer) or reject it with a comment. The manager is notified of the decision.

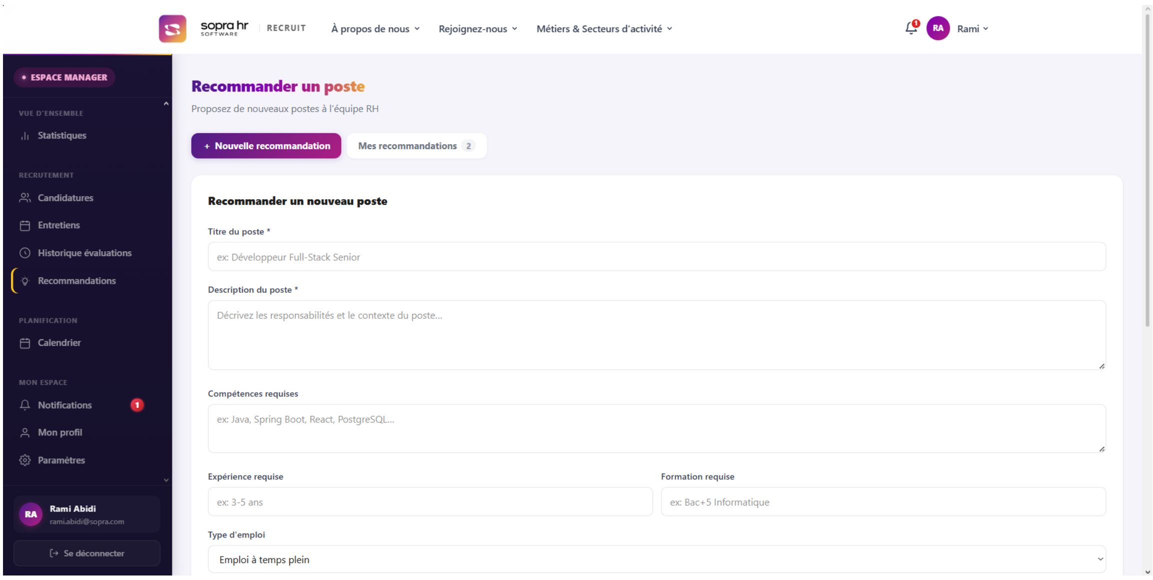

FIGURE 4.14 – Offer Recommendation Interface

Conclusion

This chapter presented the complete realization of Sprint 2 of the Sopra Recruit project. Starting from the sprint backlog and objectives, the analysis phase covered the use case diagram and detailed textual descriptions of the four main use cases : Manage Job Offer, Apply to Job Offer, Manage Application Pipeline, and Submit Offer Recommendation. The design phase introduced the sequence diagrams illustrating the interaction flows for each use case, as well as the class diagram representing the structural entities of the recruitment management module. Finally, the sprint realization section showcased the implemented user interfaces, demonstrating that all planned user stories were successfully delivered. The recruitment management layer is now fully operational and provides the functional foundation on which the AI-powered CV analysis and scoring features of Sprint 3 will be built.

Chapter

5

SPRINT 3 : AI-POWERED CV ANALYSIS AND SCORING

Plan

Introduction

Sprint 3 introduces the intelligence layer of the Sopra Recruit platform. This sprint focuses on automated CV analysis and candidate scoring to support HR decision-making and improve recruitment efficiency. The implemented features aim to provide clear, explainable, and actionable insights while preserving the existing recruitment workflow delivered in previous sprints.

5.1 Sprint Backlog

Table 5.1 presents the sprint backlog for Sprint 3, including the main user stories, estimated duration, implementation tasks, and difficulty level.

User Story	Duration	Task	Difficulty
Automatic CV analysis	1 week	Integrate AI-based CV parsing ; extract key information (skills, experience, education, languages) ; normalize extracted data into a structured candidate profile.	High
Candidate scoring engine	1 week	Implement matching logic between CV content and job requirements ; compute weighted scores ; store and expose score details.	High
Scoring criteria configuration	1 week	Build HR interface for defining and adjusting weighted scoring criteria per offer ; allow priority configuration for skills, experience, and education.	Medium
HR score visualization and ranking	4 days	Build interfaces to display score breakdown, matching criteria, and candidate ranking ; add filters and sorting by score ; implement score-based recalculation.	Medium
Analysis traceability	3 days	Provide transparent explanation fields for score contribution (skills, experience, keywords) and fallback behavior for incomplete CVs.	Medium

TABLE 5.1 – Sprint 3 Backlog

5.2 Sprint Objective

The objective of Sprint 3 is to enhance the recruitment platform with an AI-assisted evaluation workflow that automatically analyzes CVs and produces structured candidate scores.

By the end of this sprint, HR officers should be able to view ranked candidates with clear scoring indicators, compare applicants more efficiently, and make faster and more consistent pre-selection decisions.

5.3 AI Module Integration

The AI-powered CV analysis is implemented as a separate Python microservice that communicates with the main application via REST APIs. This modular approach allows independent scaling and technology flexibility.

5.3.1 Processing Pipeline

The AI module processes CVs in two steps :

1. **Text Extraction** : The system extracts text content from uploaded PDF files, with automatic fallback to OCR for scanned documents.
2. **Data Structuring** : The extracted text is analyzed to identify key sections (skills, experience, education, languages, certifications) and candidate contact information.

5.3.2 Scoring Mechanism

The scoring engine uses machine learning to measure how well a candidate's profile matches job requirements :

— Computes a global matching score (0–100) between the CV and offer criteria.
— Applies configurable weights to different criteria (skills, experience, education).

5.4 Analysis

5.4.1 Use Cases of Sprint 3

The Sprint 3 use cases define the interactions required to analyze candidate CVs, compute matching scores, and present understandable results to HR users [11].

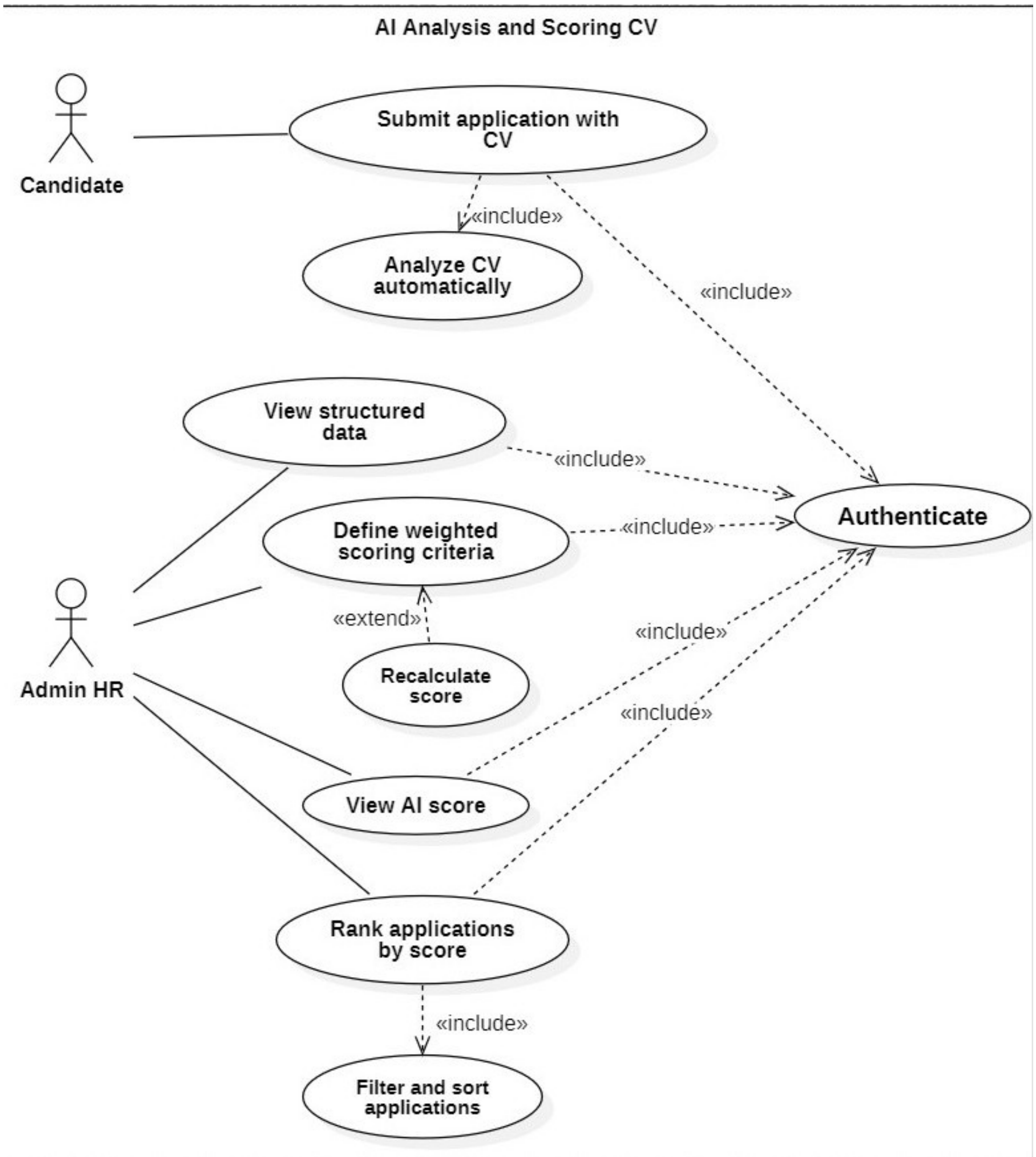

FIGURE 5.1 – Use Case Diagram — Sprint 3

Use Case : CV Analysis and Scoring	
Use Case	CV Analysis and Scoring
Initiating Actor	HR Officer (Administrator)
Goal	Automatically analyze submitted CVs and compute a matching score against a target job offer to support pre-selection.
Pre-condition	A candidate application exists with a valid CV file ; a target job offer exists with defined requirements and weighted criteria.
Post-condition	The system stores extracted CV information, the global matching score, and the detailed score breakdown ; results are available in the HR interface.
Scenario Description	The HR officer opens the applications list and triggers candidate evaluation. The system sends the CV to the AI module, which extracts text from the PDF, identifies key information (skills, experience, education, languages), and computes a matching score against job requirements. Results are stored and displayed to the HR officer with ranked candidates and score breakdowns.
Main Scenario (Success)	1. The HR officer selects a candidate application linked to a job offer. 2. The system sends the CV file to the AI module for analysis. 3. The AI module extracts text from the PDF (with OCR fallback for scanned documents). 4. The AI module identifies structured information : skills, experience, education, certifications, and languages. 5. The scoring engine computes a matching score based on job requirements and configured weights. 6. The system stores analysis results and updates candidate ranking. 7. The HR officer views the final score (0–100) and detailed breakdown.
Alternative Scenarios	**A1 — CV unreadable :** The system marks the analysis as failed and requests a new CV upload. **A2 — Incomplete CV data :** The system calculates a partial score and indicates missing information. **A3 — AI service unavailable :** The system displays an error and allows retry or manual evaluation.

TABLE 5.2 – Textual Description of Use Case "CV Analysis and Scoring"

5.4.2 Activity Diagram

This section presents the activity diagram for the main Sprint 3 user story.

5.4.2.1 Activity Diagram : CV Analysis and Scoring

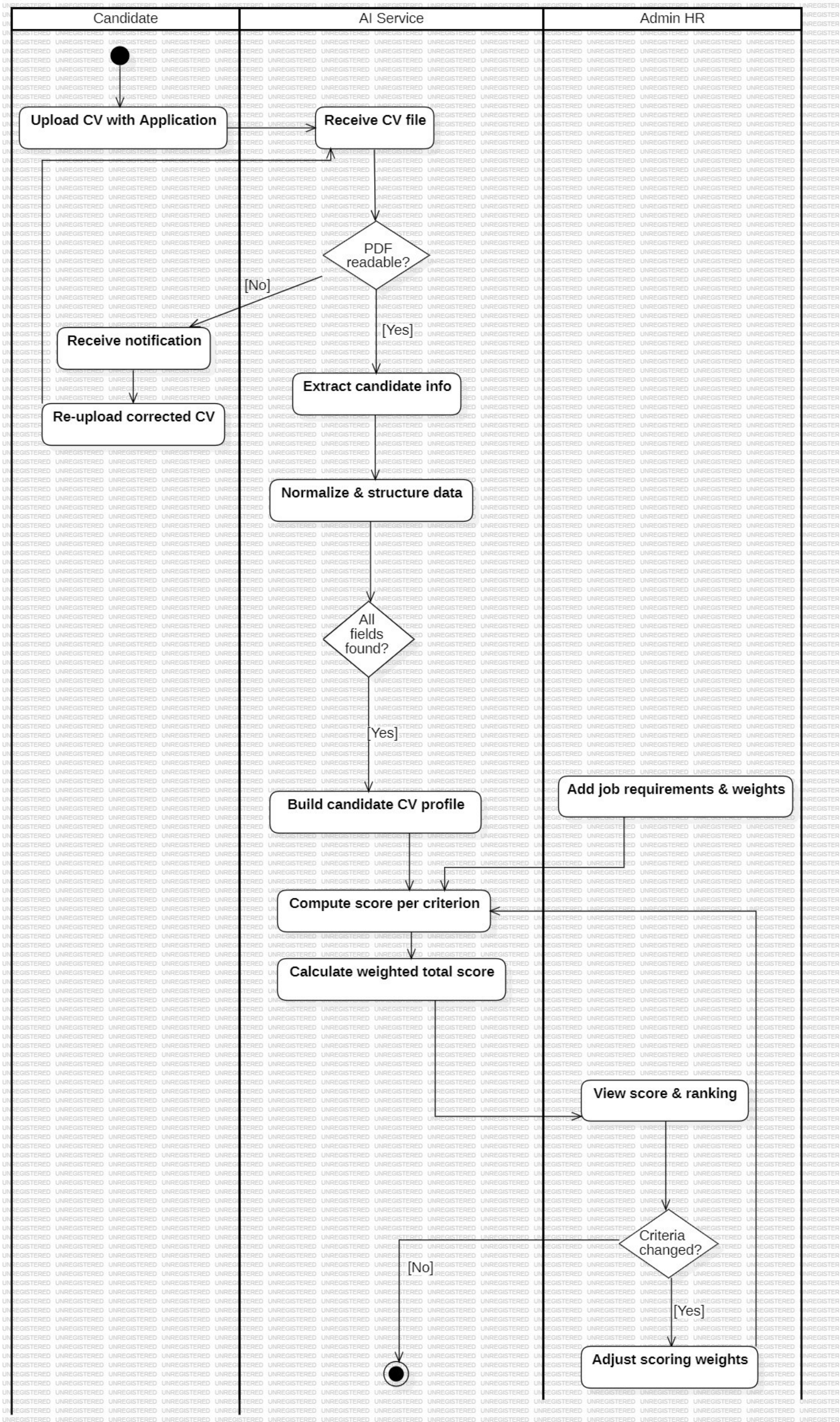

FIGURE 5.2 – Activity Diagram — CV Analysis and Scoring

5.4.3 Class Diagram

The class diagram below describes the structural entities introduced or extended in Sprint 3 for CV analysis, score computation, and results persistence [13].

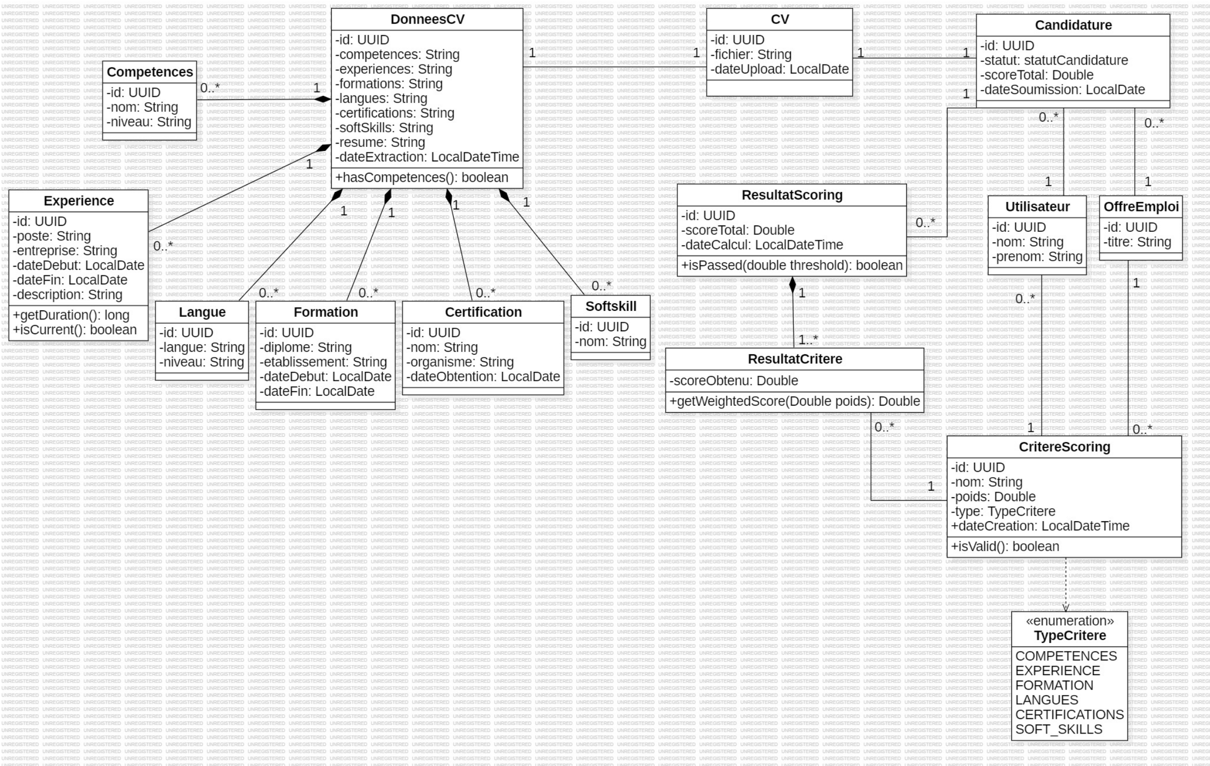

FIGURE 5.3 – Class Diagram — Sprint 3

5.5 Sprint Realization

This section presents the user interfaces implemented in Sprint 3. Image spaces are intentionally kept as placeholders and can be replaced later with final screenshots.

5.5.1 CV Analysis Result Interface (HR View)

This interface allows the HR officer to view the extracted CV data and the computed matching score for each candidate.

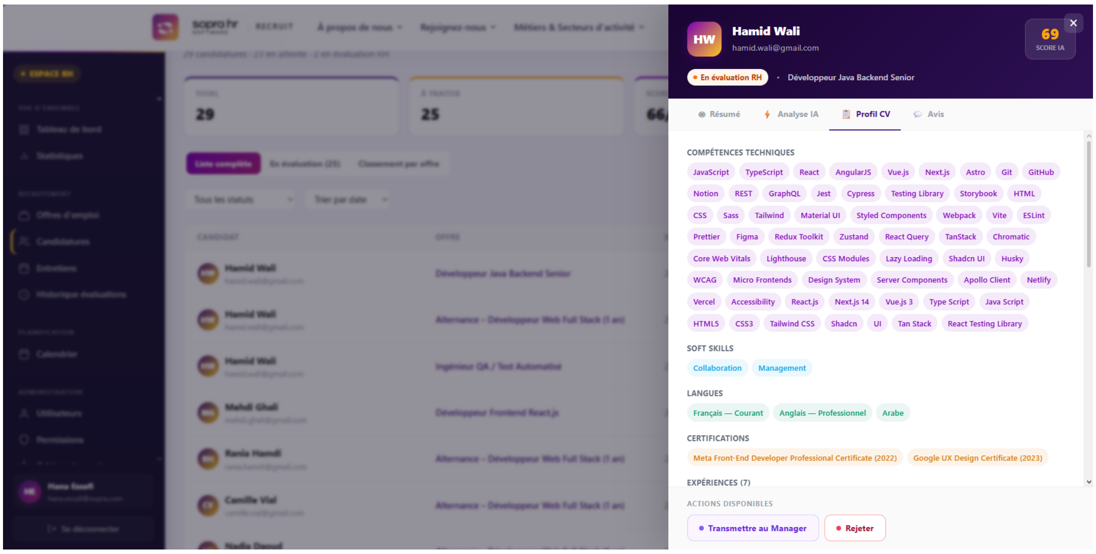

FIGURE 5.4 – CV Analysis Result Interface — HR View

5.5.2 Candidate Score Detail Interface

This interface displays the detailed score breakdown, including matching criteria such as skills, experience, and keyword relevance.

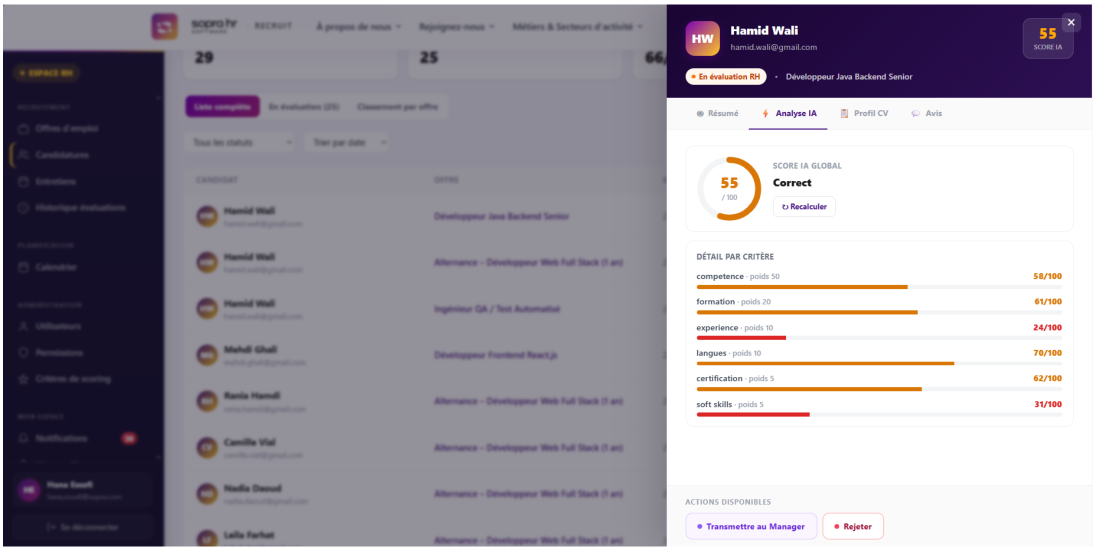

FIGURE 5.5 – Candidate Score Detail Interface

5.5.3 Candidate Ranking Interface

This interface enables HR to rank and compare applicants based on computed scores and filtering options.

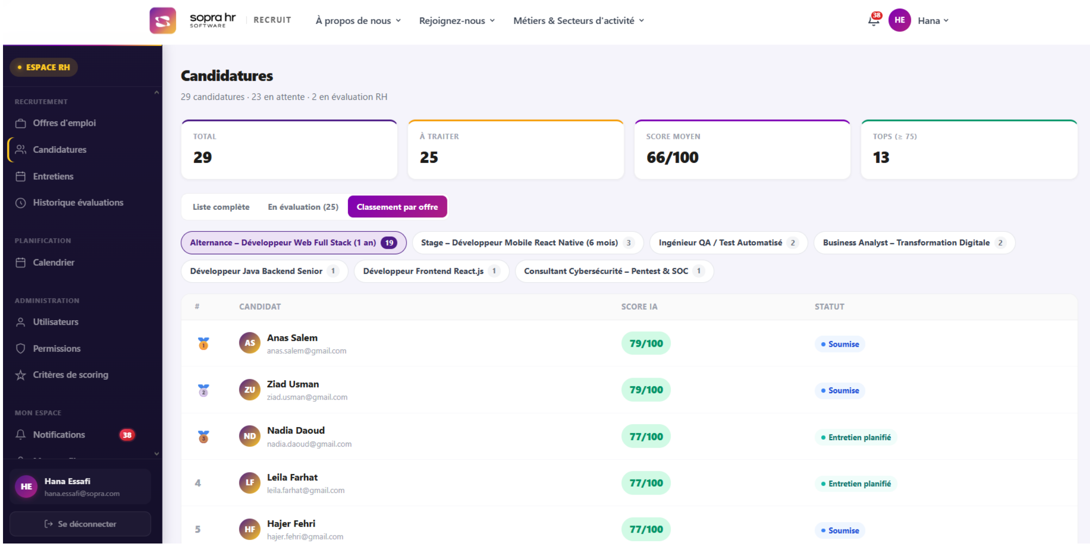

FIGURE 5.6 – Candidate Ranking Interface

Conclusion

This chapter presented the structure and realization plan of Sprint 3, focused on AI-powered CV analysis and candidate scoring. It covered the sprint backlog, objective, analysis artifacts, and realization interfaces that support HR decision-making through automated and explainable candidate evaluation.

Chapter

6

SPRINT 4 : FINALIZATION AND DEPLOYMENT

Plan

Introduction

Sprint 4 focuses on the finalization of the Sopra Recruit platform, production readiness, and operational planning. This sprint consolidates the previous increments by stabilizing the workflow, preparing deployment assets, and introducing interview planning capabilities to complete the end-to-end recruitment cycle.

6.1 Sprint Backlog

Table 6.1 summarizes the Sprint 4 backlog, including finalization tasks, deployment preparation, and interview planning.

User Story	Duration	Task	Difficulty
Interview scheduling and management	1 week	Implement interview scheduling flow (time slot selection, calendar management, validation, status update, and notification to candidates and managers).	High
Structured evaluation and decisions	1 week	Build evaluation forms with structured comments and final decision (accepted, rejected) ; implement evaluation history per application ; enable manager review workflow.	High
Notification system	1 week	Implement real-time notifications for recruitment events (application status changes, interview invitations, evaluation results) ; manage notification preferences.	Medium
Recruitment dashboard and statistics	4 days	Finalize dashboard indicators (total applications, published offers, selection rate) ; implement exportable statistics and operational tracking reports.	Medium
Deployment and finalization	3 days	Resolve pending defects, refine UI consistency, prepare environment configuration, deployment checklist, and production release procedure.	Medium

TABLE 6.1 – Sprint 4 Backlog

6.2 Sprint Objective

The objective of Sprint 4 is to deliver a stable and deployable version of the platform by finalizing key business workflows, especially interview planning, and preparing all technical and functional elements required for production use.

6.3 Analysis

6.3.1 Global Use Case Diagram

The global use case diagram consolidates the final system interactions among Candidate, HR Officer, and Manager after integrating all sprint deliverables [11].

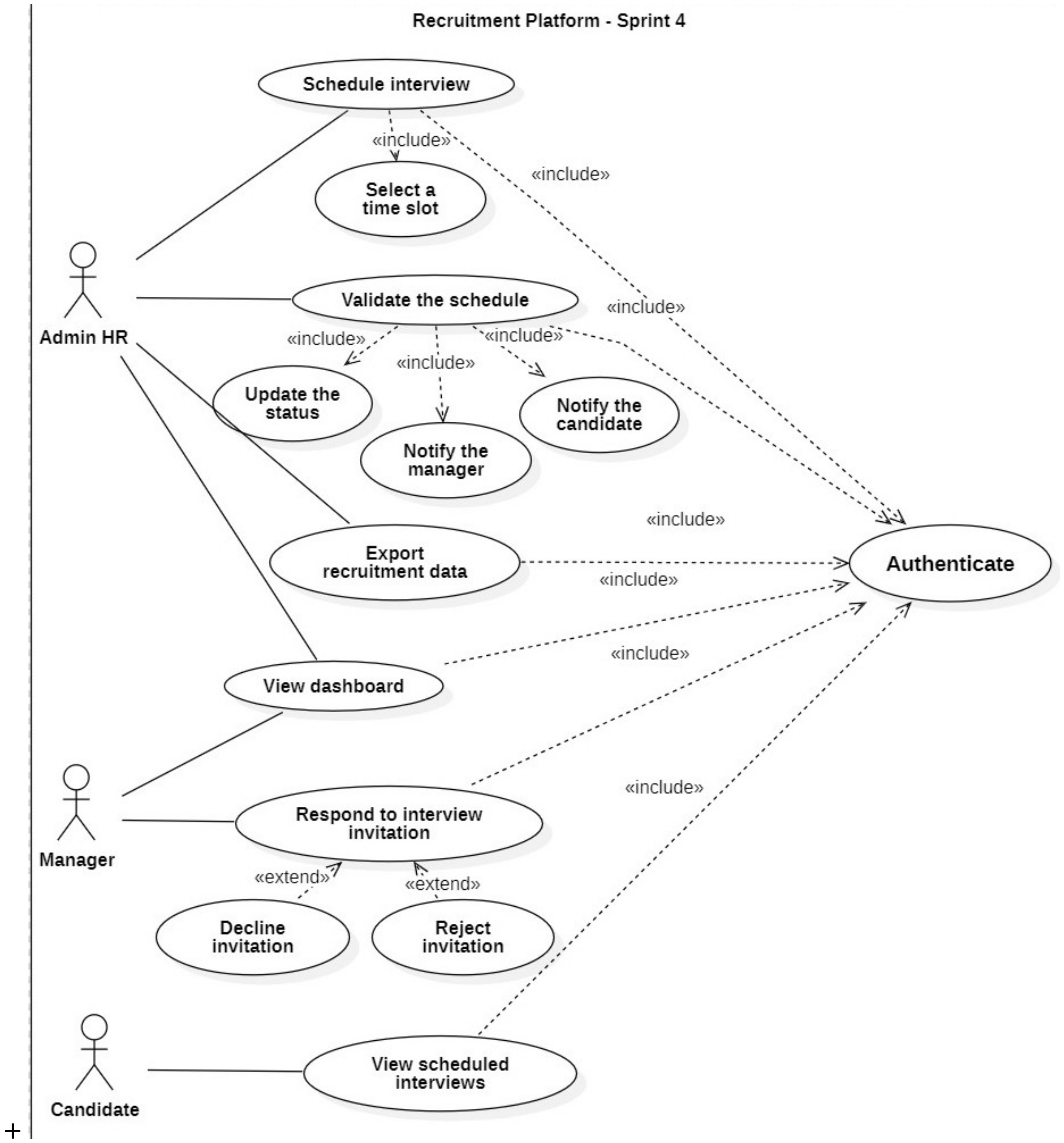

FIGURE 6.1 – Global Use Case Diagram — Final System

6.3.2 Global Class Diagram

The global class diagram presents the complete structural model of the final platform, including authentication, recruitment management, AI scoring, and interview planning entities [13].

FIGURE 6.2 – Global Class Diagram — Final System

6.3.3 Activity Diagram : Interview Planning

This activity diagram describes the end-to-end interview planning process, from selecting a candidate to confirming interview scheduling and notifications.

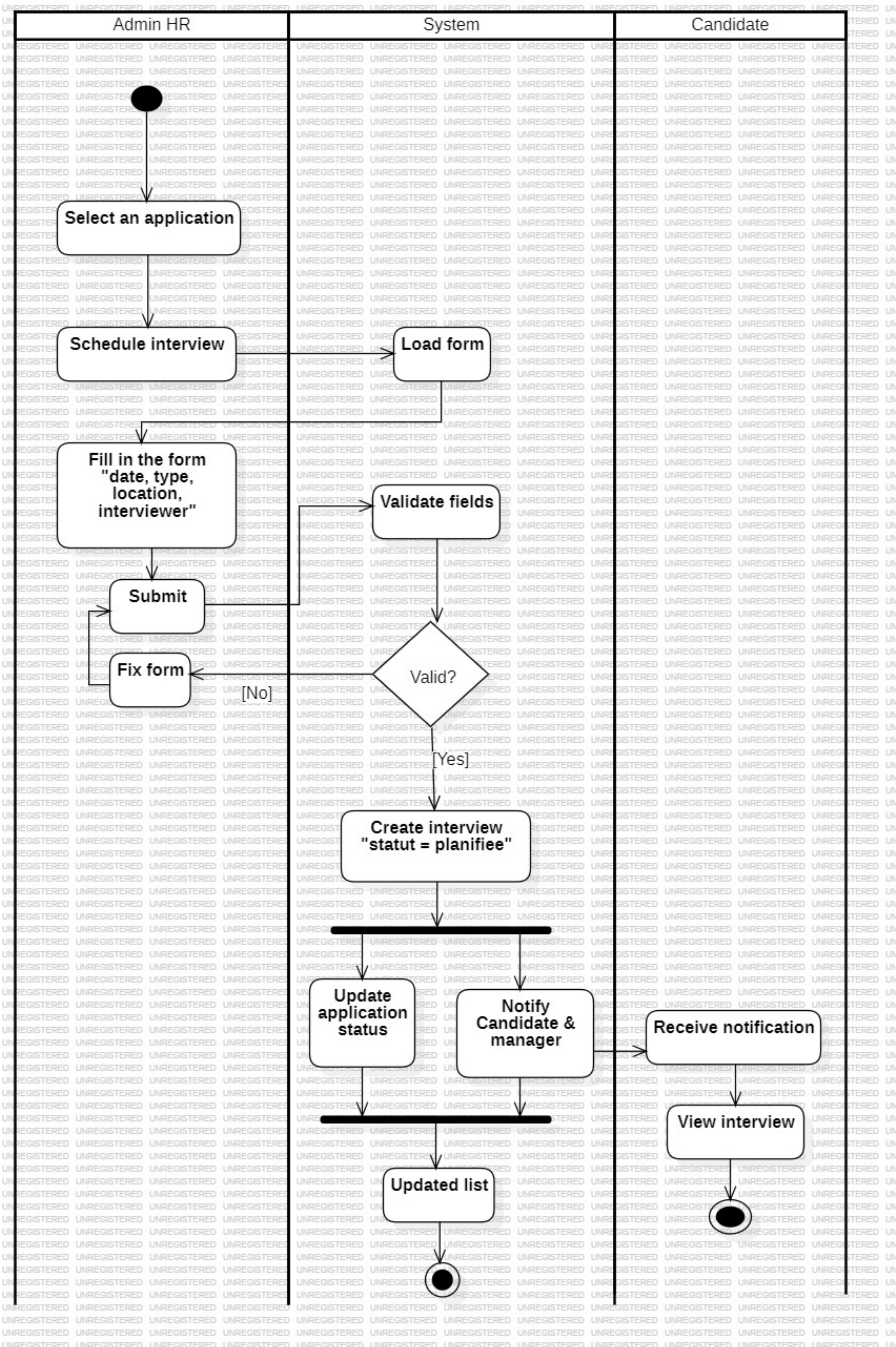

FIGURE 6.3 – Activity Diagram — Interview Planning

6.4 Sprint Realization

This section presents the final interfaces delivered in Sprint 4. Figure spaces are intentionally kept as placeholders and can be replaced later with actual screenshots.

6.4.1 Interview Scheduling Interface (HR View)

This interface allows the HR officer to schedule interviews by selecting candidate, date, time, and interviewer.

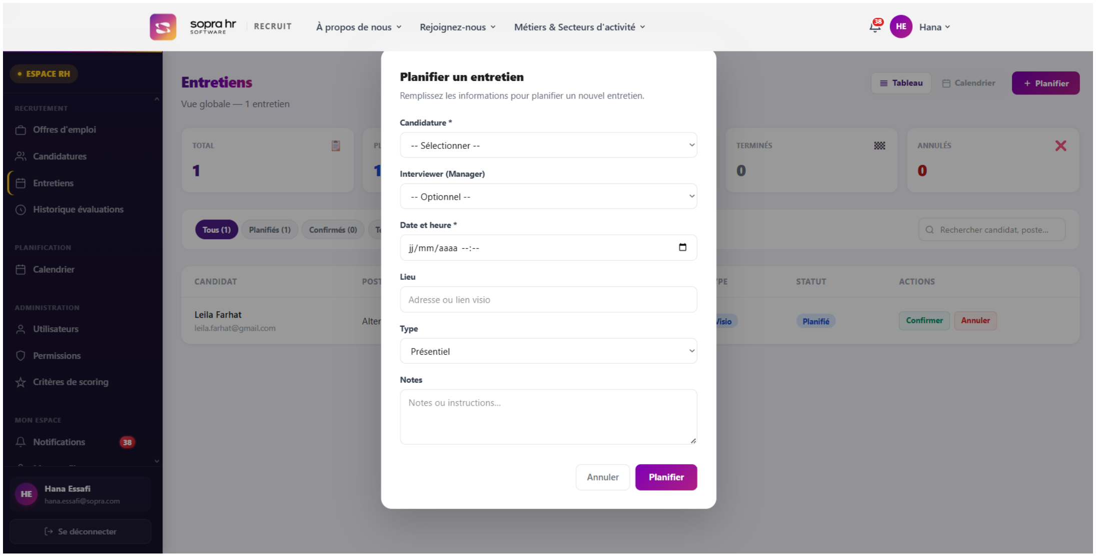

FIGURE 6.4 – Interview Scheduling Interface — HR View

6.4.2 Interview Calendar Interface

This interface displays planned interviews in calendar format and supports follow-up and updates.

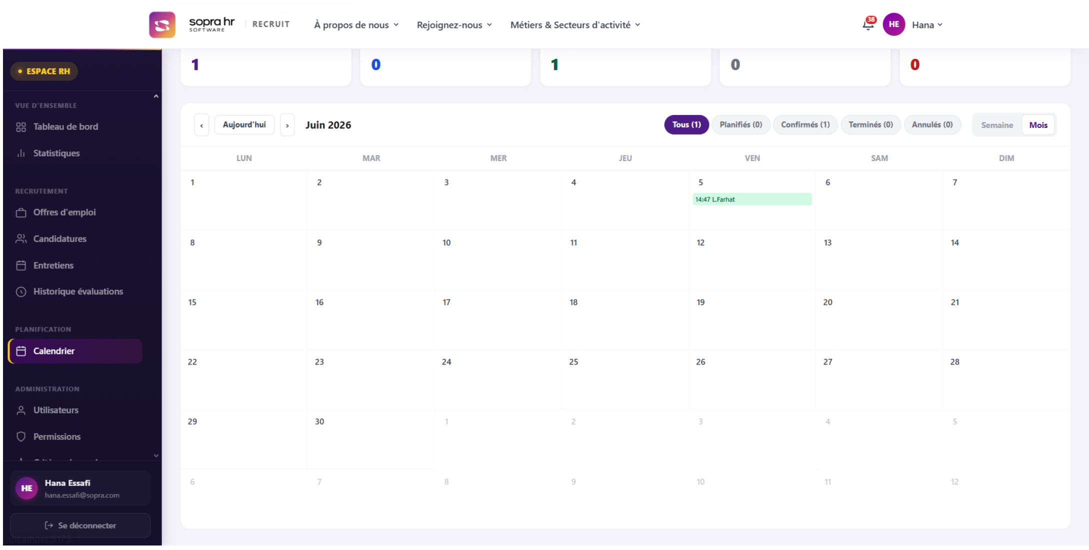

FIGURE 6.5 – Interview Calendar Interface

6.4.3 Candidate Interview Notification Interface

This interface reflects the interview notification and confirmation details visible to the candidate.

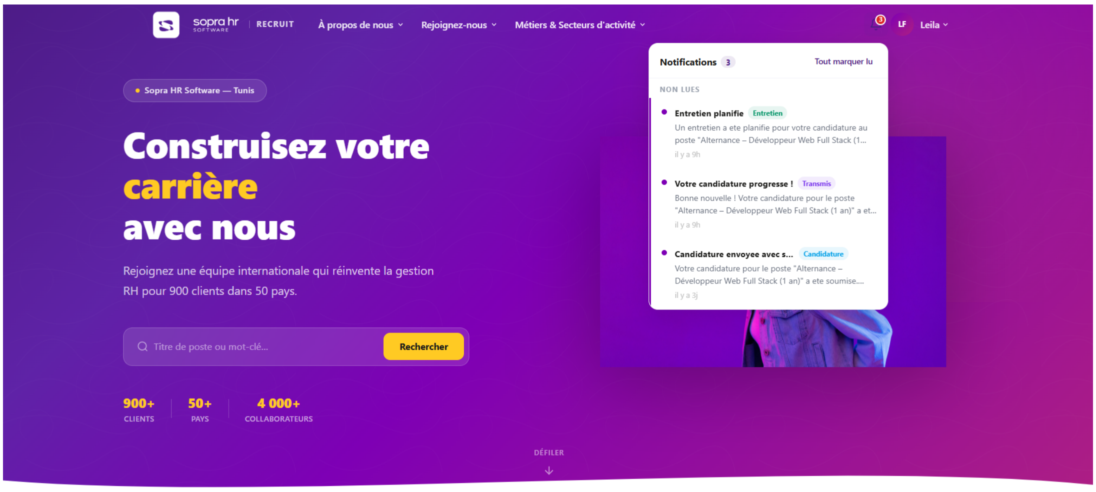

FIGURE 6.6 – Candidate Interview Notification Interface

Conclusion

Sprint 4 completed the platform by consolidating all business processes and preparing deployment. The global analysis artifacts and interview planning workflow demonstrate that the system is functionally complete, stable, and ready for operational use.

GENERAL CONCLUSION

This end-of-study project led to the design and implementation of Sopra Recruit, an intelligent recruitment portal tailored to Sopra HR Software needs. The work progressed incrementally through four sprints, covering authentication and user management, recruitment workflow management, AI-powered CV analysis and scoring, and finalization with deployment readiness and interview planning.

From a functional perspective, the platform now supports the full recruitment lifecycle : job offer publication, candidate applications, pipeline processing, manager collaboration, automated CV analysis, score-based pre-selection, and interview scheduling. From a technical perspective, the solution combines a modern web architecture, structured data flow, and explainable evaluation mechanisms that improve decision support for HR teams.

The project outcomes confirm the value of a custom-built recruitment system : reduced manual processing effort, faster candidate screening, improved traceability, and better consistency in hiring decisions. In addition, adopting an Agile Scrum approach ensured progressive delivery, continuous validation, and alignment with stakeholder priorities throughout development.

As future work, the platform can be extended with advanced analytics, multilingual CV intelligence, external calendar synchronization, and broader integration with enterprise HR ecosystems.

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Résumé

Ce projet de fin d'études porte sur la conception et le développement d'un portail de recrutement intelligent destiné à Sopra HR Software. La plateforme Sopra Recruit centralise le processus de recrutement avec la gestion des offres d'emploi, le suivi des candidatures, l'analyse automatique des CV par IA, le scoring des candidats et la planification des entretiens. Le développement suit la méthodologie Agile Scrum en quatre sprints. Les technologies utilisées sont Spring Boot, React, PostgreSQL et Python.

Mots-clés : Recrutement intelligent, Portail web, IA, Analyse de CV, Scoring, Spring Boot, React, Scrum.

Abstract

This project is about building a smart recruitment website for Sopra HR Software. The platform is called Sopra Recruit. It helps the company manage job offers, track applications, and analyze CVs using artificial intelligence. The system gives each candidate a score and helps HR plan interviews. We built this project using the Agile Scrum method in four sprints. We used Spring Boot for the server, React for the website, PostgreSQL for the database, and Python for the AI part.

Keywords : Smart recruitment, Web portal, AI, CV analysis, Scoring, Spring Boot, React, Scrum.